Aklan State University
Ibajay, Aklan

CHARRMPASS: A Web-Based Smart Entry and Exit Monitoring System Using Radio Frequency Identification (RFID) Technology for Aklan State University - Ibajay Campus

A Thesis

Presented to the Faculty of the
College of Hospitality and Rural Resource Management
Aklan State University– Ibajay Campus, Ibajay,Aklan,Philippines

s

In Partial Fulfillment of the
Requirements for the Degree

BACHELOR OF SCIENCE IN COMPUTER SCIENCE

By:
RAIAN LEE D. VALLEJO
JOHN LOUIE S. CASTILLON
JIRAH DIWATA T. AGUSTIN
RHEA MIE G. ASCAÑO

December 2025

Aklan State University
Ibajay, Aklan

APPROVAL SHEET

This thesis, "**TITLE**", with Research Code _____**,** prepared and submitted by **NAMES** in partial fulfillment of the requirements for the degree, **Bachelor of Science in Computer Science**, is hereby accepted.

REIMAR R. TINGGA, ME
Adviser

_____
Date

RODOLFO G. DOROMAL, Jr., PhD
Panel Member

_____
Date

EDWARD S. GUMBAN, MIT
Panel Member

_____
Date

JOSEPH Z. MASULA, MSCS
Panel Member

_____
Date

Accepted as partial fulfillment of the requirements for the degree, **Bachelor of Science in Computer Science.**

CYRIL D. AUSAN, MSCS
Chairperson, BSCS program

_____
Date

GEMMA F. QUINTANA, PhD
Chairperson, RDS

_____
Date

JERBY J. PADERES, MAEd
Campus Director

_____
Date

ACKNOWLEDGEMENT

In this section, express gratitude to the individuals and institutions that supported the researchers throughout your research. Begin by thanking your thesis adviser and panel members for their guidance and insights. Acknowledge your professors and mentors who contributed to your academic growth. Recognize your classmates and peers who provided assistance or encouragement during the research process. Finally, include personal acknowledgments for family and friends who offered motivation and emotional support. Keep your tone formal yet sincere, ensuring that your appreciation is clear and well-structured.

ALL THANKS TO RAIAN.

TO RAIAN ULIT.

AGAIN TO RAIAN LEE, SALAMAT.

INITIALS OF AUTHORS

(e.g. D. G. R.)

ABSTRACT

In this section, write a clear and concise summary of your thesis. Begin by introducing your research topic and stating the main problem or objective of your study. Then, briefly describe the methods you used to gather and analyze data. Summarize the key findings and explain their significance in relation to your research goals. Finally, include any important conclusions or recommendations based on your results. Keep your abstract focused and within 150–250 words, ensuring it is easy to understand while maintaining a formal and professional tone.

Keywords: *keyword, keyword, keyword*

TABLE OF CONTENTS

		Page
TITLE PAGE		i
APPROVAL SHEET		ii
ACKNOWLEDGEMENT		iii
ABSTRACT		v
TABLE OF CONTENTS		vii
LIST OF TABLES		xi
LIST OF FIGURES		xii
LIST OF APPENDICES		xiii
CHAPTER		
I	**INTRODUCTION**	1
	Background of the Study	1
	Objectives	3
	Theoretical Framework	4
	Conceptual Framework	5
	Significance of the Study	6
	Scope and Limitations of the Study	8
	Definition of Terms	9
II	**REVIEW OF RELATED LITERATURE**	11
	Topic 1 (provide topic title)	11

	Topic 2 (provide topic title)	14
	Topic 3 (provide topic title)	16
	Topic 4 (provide topic title)	21
	Topic 5 (provide topic title)	23
	Synthesis	31
III	**METHODOLOGY**	35
	Research Design	35
	System Development Process	36
	Locale of the Study	37
	Respondents of the Study	38
	Sampling Procedure	39
	Data Gathering Procedure	40
	Research Instruments	44
	Instrument Validation	48
	Data Analysis	49
	Ethical Consideration	50
IV	**PRESENTATION, ANALYSIS, AND INTERPRETATION OF DATA**	52
V	**SUMMARY, CONCLUSIONS AND RECOMMENDATIONS**	80
	Summary	80
	Findings	81
	Conclusions	83
	Recommendations	84

Areas for Further Research 85

REFERENCES 86

APPENDICES 94

LIST OF TABLES

Table		**Page**
1	Title of Table	39
2	Title of Table	53
3	Title of Table	54
4	Title of Table	57
5	Title of Table	67

LIST OF FIGURES

Figure		Page
1	Title of Figure	6
2	Title of Figure	15
3	Title of Figure	19
4	Title of Figure	20
5	Title of Figure	29

LIST OF APPENDICES

Appendix		Page
A	Letter for Permission to Conduct the Study	85
B	Consent Form	89
C	Research Instruments	90
D	Documentation	95
E	Transcripts (Qualitative)/Raw Data and Results (Experimental)	99
F	Plagiarism Check Certification	100
G		99
H		100
I		106
J		108
K		109
L		113
M		114
N		115
O		117
P		120
Q		183

CHAPTER I

INTRODUCTION

This chapter presents the background of the study. It discusses the research problem, objectives, theoretical framework, conceptual framework, significance of the study, scope and limitations, definition of terms, and other relevant information that provides the foundation for the research.

Background of the Study

Managing the entry and exit of campus is an important part of maintaining security and an organized environment in educational institutions. Entry and exit monitoring involves identifying individuals and vehicles entering and leaving the campus, verifying whether access is authorized, and keeping records of their movements. Effective monitoring helps institutions maintain awareness of who and what enters and leaves their premises and supports the safety of students, faculty, staff, and visitors. In educational settings, appropriate technologies can support efforts to improve the organization and security of campus entry and exit activities. In the Philippines, the importance of developing safe and efficient environments is also reflected in the country's long-term development goals under Ambisyon Natin 2040, which envisions a “Matatag, Maginhawa, at Panatag na Buhay” for all Filipinos (NEDA, 2016). In line with the United Nations Sustainable Development Goal (SDG) 11, which promotes safe, inclusive, resilient, and sustainable communities, maintaining organized and secure environments is an important consideration for institutions such as universities (United Nations, 2015).

The current practice at Aklan State University–Ibajay Campus involves manual monitoring of individuals and vehicles entering and leaving the campus. Security personnel are responsible for observing individuals and vehicles, checking the necessary information, and determining whether access is permitted. For visitors, relevant information is requested and recorded in a logbook or other written record, while necessary vehicle information is checked and documented. These records are maintained by security personnel for monitoring and future reference. When previous vehicle or visitor information needs to be reviewed, security personnel must manually examine the written records to locate the required details. This practice, characterized by pen-and-logbook entries, is inefficient, prone to errors, and time-consuming, particularly when entry and exit information must be consistently documented and retrieved.

Aklan State University–Ibajay Campus is an educational institution where students, faculty, staff, visitors, and vehicles enter and leave the campus as part of daily activities. Within the campus, security personnel are responsible for monitoring and recording entry and exit activities. Based on preliminary observation, the current process involves checking and recording the necessary information of people and vehicles entering and leaving the campus using a logbook. The process involves security personnel as the primary monitors, while students, faculty, staff, visitors, and vehicle users are among those affected by the monitoring process.

The entry and exit monitoring process at Aklan State University–Ibajay Campus presents difficulties in consistently checking individual and vehicle information and maintaining organized entry and exit records. The researchers also observed difficulties in consistently checking individual and vehicle information and maintaining organized entry and exit records. The process relies mainly on analog methods for checking and recording, which can make the identification and verification of individuals and vehicles less consistent. Security personnel may also encounter difficulties when monitoring students, faculty, staff, visitors, and vehicles entering and leaving the campus, particularly when information needs to be checked and recorded during entry and exit activities. In addition, the lack of a centralized method for managing user information, vehicle information, and access records may make it difficult to maintain organized records for later review. These limitations indicate areas in the current monitoring process that need to be addressed to support more systematic user and vehicle identification, access verification, and entry and exit record management.

Based on the researchers' preliminary observation and interview among 150 respondents composed of one hundred forty one (141) students, seven (7) faculties, (1) university student council and (1) security personnel…… at Aklan State University–Ibajay Campus, the checking of individuals and vehicles entering and leaving the campus is conducted through existing gate procedures, with particular attention given to visitors and vehicles not associated with the university. The researchers also observed that individual and vehicle information is not checked consistently during entry and exit activities. The preliminary interview further confirmed the current reliance on manual checking and

handwritten recording of information. During the preliminary interview, security personnel reported that the same logbook is used to record information about individuals and vehicles entering and leaving the campus. These observations and interview findings provide local evidence that the existing campus entry and exit monitoring process relies on manual procedures and does not consistently verify individual and vehicle information. Supporting this observation, Mimbala et al. (2024) reported that Mindanao State University–Lanao del Norte Agricultural College relied on manual pen-and-logbook procedures for monitoring vehicle entry and exit, with the researchers identifying issues involving manual monitoring, inaccurate recording, and delays in providing reports.

The identified limitations may affect the efficiency of monitoring activities at the campus entrance and exit. Based on the researchers' preliminary observation and interview, when individual and vehicle identification and verification depend mainly on analog technology, such as visual observation and handwritten recording of vehicle and visitor information, security personnel may need to spend additional effort checking vehicles and visitors and maintaining or reviewing records. The lack of centralized entry and exit information may also make it difficult to retrieve and track previous vehicle and visitor activities when such information is needed. As a result, monitoring individual and vehicle movements and determining which vehicles and visitors have entered or exited the campus can become less organized and less efficient. These effects highlight the need for a more systematic approach to vehicle identification, verification, visitor monitoring, and record management.

Although existing studies and implementations have demonstrated the use of digital technologies for campus entry and exit monitoring, these systems have been developed according to the specific users, processes, and operational requirements of their respective institutions. Wahi (2024), for example, examined an RFID-integrated monitoring system at Politeknik Mukah Sarawak to address challenges in student entry and exit monitoring, particularly those associated with traditional manual monitoring and record-keeping. Similarly, Jacaban et al. (2023) developed a Campus Entry and Exit Monitoring System at Don Mariano Marcos Memorial State University–South La Union Campus using Quick Response (QR) codes to digitally record student entry and exit information. Wahab et al. (2025) also developed an RFID-based student entry and exit monitoring system at University College TATI to address limitations in manual gatekeeping and improve the recording of student movements. Morpos et al. (2025) developed SMART ACCESS at Eastern Visayas State University–Ormoc Campus, which uses Radio Frequency Identification (RFID), a web-based interface, and a centralized database for gate entry and exit verification of students, employees, vendors, and visitors; however, the study reported verification delays associated with internet connectivity. In addition, Ateneo de Manila University implemented an RFID access system in which students and employees use RFID-enabled identification cards to enter and exit university premises, while vehicles with RFID tags use designated RFID lanes; the university also planned additional Closed-Circuit Television (CCTV) cameras to support campus security and access verification. Similarly, Manuel S. Enverga University Foundation (MSEUF) implemented RFID vehicle tags with boom barrier access to provide automated entry and exit for authorized four-wheel vehicles. These existing

implementations demonstrate that RFID and other digital technologies can support identification, access verification, and entry and exit monitoring; however, they differ in their target users, technology, monitoring scope, and institutional requirements. Meanwhile, the current process at Aklan State University–Ibajay Campus (ASU–Ibajay) still relies mainly on manual checking and handwritten recording of information concerning individuals and vehicles entering and leaving the campus. Such a process does not provide a centralized means of electronically identifying authorized campus users and vehicles, verifying access, and systematically organizing entry and exit records for monitoring and later reference. Therefore, an unaddressed need remains for a systematic and centralized entry and exit monitoring approach suited to the operational requirements of ASU–Ibajay and capable of accommodating both campus users and vehicles.

In response to the identified problem and gap, this study proposes the development of CHARRMPASS: A Web-Based Smart Entry and Exit Monitoring System Using Radio Frequency Identification (RFID) Technology for Aklan State University–Ibajay Campus. The proposed system will use Radio Frequency Identification (RFID) technology to identify campus users and registered vehicles and record their entry and exit transactions. Through a web-based platform, authorized personnel will be able to access and manage campus user information, vehicle information and entry and exit records through a centralized system. CHARRMPASS is intended to reduce dependence on manual identification and record-keeping by providing a more structured approach to campus entry and exit monitoring. Through this approach, the study aims to support more

organized, efficient, accurate, and secure monitoring of the entry and exit activities of people and vehicles within the campus.

Objectives of the Study

This study aimed to develop CHARRMPASS: A Web-Based Smart Entry and Exit Monitoring System Using Radio Frequency Identification (RFID) Technology for Aklan State University–Ibajay Campus. The system was intended for students, faculty, staff, visitors, and authorized security personnel to support campus security and promote a more efficient, organized, and systematic process for monitoring the entry and exit of individuals and vehicles.

Specifically, the study aimed to:

1. Develop the following system modules:

    1.1 Registration and Permit Module – Allows students, faculty, and staff to register their personal and vehicle information and associate their registered RFID card or tag with their records.

    1.2 RFID Identification and Access Module – Enables automated scanning and validation of RFID-enabled individuals and vehicles at the campus entry and exit points to verify authorized access and support secure identification.

    1.3 Centralized Web Dashboard and Monitoring Module – Provides security personnel with a web-based interface to monitor entry and exit

activities, manage registered users, and view real-time and historical records of individual and vehicle transactions.

1.4 Visitor Management Module – Allows the temporary registration and monitoring of visitors and their vehicles through manual registration or temporary RFID identification, ensuring that visitor entry and exit activities are properly recorded.

1.5 Emergency Exemption Module – Provides priority access for authorized emergency units during urgent situations while maintaining records of their entry and exit transactions for monitoring and accountability.

1.6 Reporting Module – Allows authorized personnel to view, filter, and generate organized reports of individual and vehicle entry and exit transactions within the campus, providing accessible records for monitoring and review.

1.7 Offline Module – Allows the system to temporarily record and store individual and vehicle entry and exit transactions during network or internet interruptions, ensuring continuous monitoring at the gates and synchronizing stored records once the connection is restored.

2. Evaluate the developed system using the ISO 25010 Software Product Quality Model, specifically utilizing the following criteria:

2.1 **Functional Suitability:** This characteristic represents the degree to which a product or system provides functions that meet stated and implied needs when used under specified conditions.

2.2 **Performance Efficiency:** This characteristic represents the degree to which a product performs its functions within specified time and throughput parameters and is efficient in the use of resources (such as CPU, memory, storage, network devices, energy, materials...) under specified conditions;

2.3 **Reliability:** Degree to which a system, product or component performs specified functions under specified conditions for a specified period of time;

2.4 **Security:** Degree to which a product or system defends against attack patterns by malicious acts and protects information and data so that persons or other products or systems have the degree of data access appropriate to their types and levels of authorization. ;

2.5 **Maintainability:** This characteristic represents the degree of effectiveness and efficiency with which a product or system can be modified to improve it, correct it or adapt it to changes in environment and requirements.

Theoretical Framework

This study was anchored on General Systems Theory, Automation Theory, Database Principles, Information Security Principles, Human–Computer Interaction (HCI) Principles, Usability Principles, and Agile Software Development Principles, which provides the theoretical basis for the design and development of the CHARRMPASS system. The theory guided the researchers in understanding the interrelationship among the different components of the system and designing an integrated system in which vehicle information, Radio Frequency Identification (RFID) identification, access verification, monitoring, and data management work together to achieve the intended objectives.

General Systems Theory

General Systems Theory views a system as a unified whole composed of interacting or interdependent components. It emphasizes the relationships among the components of a system and considers how inputs are transformed through processes to produce outputs. The theory focuses on the overall behavior of the system rather than only on its individual components (von Bertalanffy, 1968).

Automation Theory

Automation Theory focuses on the use of technology and control mechanisms to perform tasks or processes with reduced human intervention. It involves the automatic operation and control of processes that may otherwise require manual effort (Groover, 2015).

The theory highlights how automated systems can improve operational performance by minimizing human error, increasing processing speed, and providing consistent results in repetitive tasks. Through RFID-based identification and automated access monitoring, CHARRMPASS supports faster and more organized verification of individuals and vehicles while helping security personnel manage and monitor campus access more efficiently.

Database Principles

Database Management Principles refer to the principles and practices used to organize, store, retrieve, update, and maintain data in a structured database. These principles support efficient data management, consistency, integrity, and controlled access to information (Elmasri & Navathe, 2016).

In CHARRMPASS, a centralized database stores individual profiles, vehicle records, RFID information, and entry and exit logs, enabling organized monitoring and reliable reporting. The system utilizes a backend platform secured through Row Level Security (RLS), JSON Web Token (JWT)-based authentication, and Role-Based Access Control (RBAC). In addition, communications between the system and its users are

secured through Hypertext Transfer Protocol Secure (HTTPS) to help protect data integrity and prevent unauthorized access.

To support automated entry and exit monitoring, the proposed system integrates an RFID-Based Verification Algorithm for identifying and verifying registered individuals and vehicles. The system compares the scanned RFID information with the records stored in the centralized database and processes the transaction based on predefined verification conditions. These conditions may include checking whether the RFID identifier is registered and active, whether the associated individual or vehicle is authorized, and whether the record meets the required entry or exit conditions. Based on the verification result, the system automatically records the transaction and identifies whether the access attempt is authorized or unauthorized.

The system also applies Role-Based Access Control (RBAC) to secure the web-based management dashboard by assigning access permissions according to the user's role, such as Guard or Administrator. While the RFID verification process uses predefined rules for automated decision-making, RBAC controls which system functions and information can be accessed by each authorized user. These automated verification and access-control mechanisms reduce manual processing, minimize recording errors, improve response time, and support reliable monitoring of individual and vehicle entry and exit activities.

Human–Computer Interaction Principles

Human-Computer Interaction (HCI) is a multidisciplinary field concerned with the design, evaluation, and implementation of interactive computing systems for human use, focusing on user experience, system accessibility, and intuitive interaction design (Dix et al., 2004).

In CHARRMPASS, HCI principles are applied to the design of the web-based interface used by authorized personnel to manage individual and vehicle information, monitor campus entry and exit activities, and access recorded transactions. The system is designed to provide clear and organized information, allowing users to interact with its functions efficiently and understand access records with minimal difficulty. By applying HCI principles, CHARRMPASS aims to provide an accessible, user-friendly, and efficient interface that supports security personnel and authorized users in performing campus access monitoring tasks.

Usability Principles

Usability Principles provide a foundation for developing systems that users can operate effectively, efficiently, and with ease. Usability considers how easily users can understand, learn, and use a system to accomplish specific tasks (ISO, 2018).

Usability principles are relevant to CHARRMPASS because the system will be used by individuals with different roles and responsibilities, particularly administrators and security personnel who need to perform vehicle registration, access verification, monitoring, and record management. The researchers applied usability principles in

organizing system functions, simplifying user interactions, and presenting relevant information clearly. This supports efficient use of the system and contributes to the overall quality of the proposed system.

Agile Software Development Principles

Agile Software Development Principles refer to an approach to software development that emphasizes iterative development, continuous feedback, collaboration, and the ability to respond to changing requirements. Software is developed and improved through repeated cycles, allowing developers to incorporate feedback and make necessary refinements throughout the development process (Beck et al., 2001).

The researchers applied Agile principles by developing the system components incrementally, testing developed functions, incorporating feedback, and refining the system throughout the development process. The approach supports the progressive development of the system's modules and allows identified issues and user requirements to be addressed during successive development activities. This foundation therefore provides support for the planning, development, testing, deployment, and improvement of CHARRMPASS.

Conceptual Framework

The conceptual framework of the study is presented in Figure 1 below. It provides a visual representation of the inputs, processes, and outputs involved in the development and evaluation of the proposed system.

Figure 1. *The Conceptual Framework of the Study*

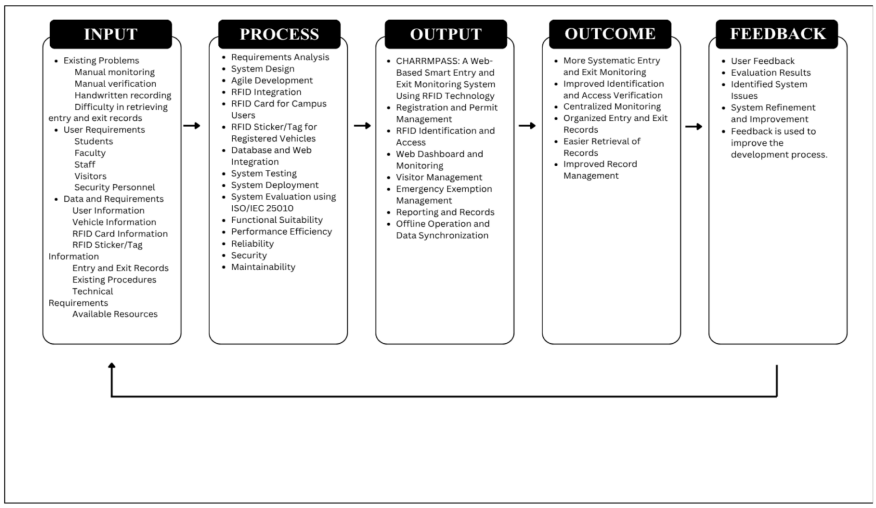


Figure 1

As shown in Figure 1, the conceptual framework of the study illustrates the relationship among the **Input, Process, Output, Outcome, and Feedback** involved in the development and evaluation of CHARRMPASS: A Web-Based Smart Entry and Exit Monitoring System Using Radio Frequency Identification (RFID) Technology for Aklan State University–Ibajay Campus, Aklan. The framework follows the **Input–Process–Output–Outcome (IPOO) model with a feedback mechanism** to present the systematic flow of the study.

The **Input** consists of the existing problems, user requirements, data, technical requirements, existing procedures, and available resources needed for the development of the system. These include manual monitoring and verification, handwritten records, user information, vehicle information, RFID card and sticker/tag information, and entry and exit records from the intended users and stakeholders.

The **Process** involves requirements analysis, system design, Agile development, RFID integration, database and web integration, system testing, deployment, and evaluation. RFID technology is used to identify authorized campus users through **RFID cards** and registered vehicles through **RFID stickers/tags**. The developed system is evaluated using selected characteristics of the **ISO/IEC 25010 Software Product Quality Model**, namely Functional Suitability, Performance Efficiency, Reliability, Security, and Maintainability.

The **Output** is the developed **CHARRMPASS** system, which includes Registration and Permit Management, RFID Identification and Access, Web Dashboard and Monitoring, Visitor Management, Emergency Exemption Management, Reporting and Records, and Offline Operation and Data Synchronization.

The **Outcome** represents the expected contribution of the system to campus entry and exit monitoring, including more systematic monitoring, improved identification and access verification, centralized monitoring, organized records, and easier retrieval of entry and exit information.

The **Feedback** represents the information obtained from user feedback, system testing, evaluation results, and identified system issues. This feedback is used to identify areas for refinement and improvement, supporting the iterative development and continuous improvement of CHARRMPASS.

Significance of the Study

This study developed a Web-Based Smart Entry and Exit Monitoring System Using Radio Frequency Identification (RFID) Technology for Aklan State University–Ibajay Campus as a basis for improving the monitoring, verification, and recording of individuals and vehicles entering and exiting the campus. The developed system was designed to address the identified needs and challenges related to the manual monitoring, verification, and recording of entry and exit activities at Aklan State University–Ibajay Campus, Aklan. The findings and output of this study may provide valuable benefits to the following:

Students, Faculty, and Staff. The developed system may help students, faculty, and staff experience a more convenient and organized entry and exit process through RFID-based identification and faster verification of authorized users and vehicles.

Security Personnel. The system may assist security personnel in monitoring and verifying individuals and vehicles entering and exiting the campus through RFID-based identification, access verification, and centralized recording of entry and exit activities.

Aklan State University–Ibajay Campus. The study may provide the university with a more organized and efficient approach to entry and exit monitoring through improved accuracy of records, efficient monitoring, and easier retrieval and management of entry and exit information.

Community. The developed system may contribute to a safer and more organized campus environment by improving the monitoring and recording of individuals and vehicles entering and exiting Aklan State University–Ibajay Campus.

Visitors. The developed system may contribute to a more organized visitor entry and exit process through the recording of visitor and vehicle information and the use of temporary RFID identification when applicable.

Future Researchers. The study may provide a useful reference for future researchers interested in RFID technology, entry and exit monitoring, web-based system development, database management, and hardware-software integration.

Scope and Limitations of the Study

This study will focus on the design, development, implementation, and evaluation of CHARRMPASS: A Web-Based Smart Entry and Exit Monitoring System Using Radio Frequency Identification (RFID) Technology for students, faculty, staff, visitors, and security personnel in Aklan State University–Ibajay Campus. It will cover the development of the identified system modules, including Registration and Permit Module, Radio Frequency Identification (RFID) and Access Module, Centralized Web

Dashboard and Monitoring Module, Visitor Management Module, Reporting Module, Offline Module, and Emergency Exemption Module. The study will also include the testing and evaluation of the developed system using the **ISO/IEC 25010 Software Product Quality Model,** particularly in terms of Functional Suitability, Performance Efficiency, Reliability, Security, and Maintainability.

This study was limited to selected students, faculty, staff, and security personnel of Aklan State University–Ibajay Campus and the specific processes and functions identified within the scope of the proposed system. It did not cover mobile application integration, GPS tracking, advanced artificial intelligence, the management, allocation, and monitoring of physical parking spaces inside the campus, or the identification and counting of the number of people inside a vehicle. The evaluation of the system was also limited to the selected ISO/IEC 25010 quality criteria, namely Functional Suitability, Performance Efficiency, Reliability, Security, and Maintainability, and the responses of selected students, faculty, staff, and security personnel who met the established inclusion criteria during the system evaluation period. Other factors beyond the defined scope of the study were not included.

Definition of Terms

For the purpose of clarity and precision, important terms used in this study are defined to provide a clear understanding of their meanings within the context of the research. The terms are defined both conceptually and operationally.

The conceptual and operational definitions of the terms used in this study are as follows:

Access Control. Refers to the process of regulating and restricting access to a controlled area through authentication and verification mechanisms (Poh & Ngadengon, 2021).

In this study, Access Control refers to the CHARRMPASS function that verifies registered user and vehicle information through Radio Frequency Identification (RFID) technology and allows authorized individuals and vehicles to enter and exit the campus.

Administrator. Refers to an authorized individual responsible for managing and overseeing system operations, including user accounts and system records (Grepon et al., 2021).

In this study, Administrator refers to the authorized user role responsible for managing CHARRMPASS data, individual accounts, vehicle records, and system monitoring through the web-based dashboard.

Centralized Database. Refers to a system that stores and manages data in a single location, allowing authorized users to access shared information efficiently (Erickson, 2025).

In this study, Centralized Database refers to the storage system of CHARRMPASS where user information, vehicle records, Radio Frequency Identification (RFID) data, and entry and exit transactions are stored and managed.

Emergency. Refers to a situation that requires immediate response due to threats to life, safety, or property, such as accidents, fires, or security-related incidents (Microsoft Learn, 2023).

In this study, Emergency refers to situations in which authorized emergency

vehicles are given priority access to enter and exit the campus when immediate response is required.

Entry and Exit Monitoring. Refers to the process of tracking and recording individuals and vehicles entering and leaving a controlled area. (Dahua Technology, n.d.).

In this study, Entry and Exit Monitoring refers to the process of identifying and recording the entry and exit activities of individuals and vehicles at the campus entry and exit points through CHARRMPASS.

Passive LF Radio Frequency Identification (RFID) Tag. Refers to a passive Radio Frequency Identification (RFID) tag that operates within the low-frequency range and does not require an internal battery because it receives power from the RFID reader. (Le Breton et al., 2021).

In this study, Passive LF RFID Tag refers to the RFID tag embedded in the card used by registered individuals for identification and verification during campus entry and exit.

RFID (Radio Frequency Identification). Refers to a wireless technology that uses radio-frequency signals to automatically identify objects through tags and readers (Rouse, 2022).

In this study, Radio Frequency Identification (RFID) refers to the core technology used by CHARRMPASS to identify registered individuals and vehicles and record their entry and exit transactions.

RFID Reader. Refers to a hardware device that detects and reads identification information from Radio Frequency Identification (RFID) tags (Rouse, 2022).

In this study, Radio Frequency Identification (RFID) Reader refers to the hardware device used by CHARRMPASS to detect RFID tags assigned to individuals and vehicles, using LF and UHF RFID readers for identification, verification, and entry and exit logging.

UHF Radio Frequency Identification(RFID). Refers to an electronic identification technology that operates within the **860–960 MHz** frequency range and uses radio waves to automatically identify tagged objects or individuals from a distance. (Laheurte et al., 2014).

In this study, UHF RFID refers to the RFID technology used to identify and verify registered individuals and vehicles during campus entry and exit

User. Refers to individuals who interact with or are registered in a system, including authorized personnel and other system users (Microsoft Learn, 2022).

In this study, User refers to individuals registered in CHARRMPASS whose information is recorded and managed by the system, including students, faculty, and staff, as well as authorized personnel who interact with the system.

Vehicle Monitoring. Refers to the systematic observation and recording of vehicle movements within a particular location (NSW Government, n.d.).

In this study, Vehicle Monitoring refers to the process of identifying vehicles and recording their movements entering and exiting Aklan State University–Ibajay Campus.

Vehicle Registration. Refers to the process of enrolling a vehicle into a system and linking it to a user and an identification tag (Kianpisheh et al., 2021).

In this study, Vehicle Registration refers to the process by which students, faculty, and staff who have vehicles register their personal and vehicle information and associate the registered vehicle with an assigned RFID tag for campus entry and exit monitoring.

Visitor. Refers to a person who is not regularly registered or affiliated with an organization and is temporarily granted access to a controlled area. (Mondal et al., 2021).

In this study, Visitor refers to a person who is temporarily registered or recorded in CHARRMPASS when entering and exiting the campus, including the necessary information for monitoring purposes.

CHAPTER II

REVIEW OF RELATED LITERATURE

This chapter presents a review and discussion of relevant literature and previous studies that provide a theoretical, conceptual, and empirical foundation for the present study. It includes related local and foreign literature and studies relevant to the development, implementation, and evaluation of the proposed system. The reviewed studies provide insights into the concepts, technologies, methods, and approaches related to RFID technology, smart vehicle access control, real-time monitoring, and security dashboards.

Chapter II is divided into several topics: **Topic A Radio Frequency Identification (RFID) Technology, (2) Smart Vehicle Access Control System, (3) Real-Time Monitoring and Security Dashboard, (4) Campus Entry and Exit Monitoring, and (5) Synthesis**. Each topic presents relevant studies that provide insights into the technologies and approaches related to the present study. The reviewed studies are analyzed to establish their relevance to CHARRMPASS and to identify gaps that the present study seeks to address.

Topic A

Radio Frequency Identification (RFID) Technology

RFID (Radio Frequency Identification) is a form of wireless communication that uses electromagnetic or electrostatic coupling in the radio frequency portion of the electromagnetic spectrum to identify an object, animal, or person uniquely.

Automatic Vehicle Access System Using Ultra-High Frequency Radio Frequency Identification (RFID)

Poh and Ngadengon (2021) developed an Automatic Vehicle Access System using Ultra-High Frequency (UHF) Radio Frequency Identification (RFID) technology to provide fast, accurate, and secure monitoring of vehicle entry and exit. The study aimed to address the difficulties of manually managing vehicle access, where some vehicles may only be visually checked and not properly recorded, allowing unauthorized vehicles to potentially enter or leave undetected. The system was designed to automatically identify registered vehicles using passive UHF Radio Frequency Identification (RFID) tags while integrating a graphical user interface (GUI) and vehicle user database for access monitoring and management.

The system integrates a passive UHF RFID tag, UHF RFID reader, Arduino Mega 2560 microcontroller, servo motor, barrier gate, GUI, and MySQL database. When a vehicle carrying a registered Radio Frequency Identification (RFID) tag enters the reader's detection range, the tag data is transferred to the system, triggering the servo motor to open the barrier gate while the vehicle information is stored in the database and displayed through the GUI. The system also allows administrators to register and edit user information, monitor scanned vehicle data, and access reports containing real-time date and time records of vehicles passing through the entrance. Field testing showed stable tag detection at distances of up to seven meters under the tested configuration, while vehicle speed, tag placement, reader angle, and windshield tint affected system performance.

However, the system mainly uses a desktop-based GUI for monitoring and management rather than a fully web-based platform that can provide centralized remote access across different devices and locations. Although the study records vehicle data and integrates it with a database, its implementation is primarily focused on automated vehicle access at a specific entrance prototype. The study itself also recommends further improvement by linking registered vehicle data with the official premises database, storing data in an off-site cloud location, and expanding the GUI to support more than one Radio Frequency Identification (RFID) reader. These limitations show that while the study effectively demonstrates passive UHF Radio Frequency Identification (RFID) for contactless vehicle identification, read-range evaluation, automated barrier gate triggering, and hardware-software integration, further development is needed toward a more centralized web-based system capable of supporting broader institutional vehicle entry and exit monitoring with real-time tracking.

Capture-Aware Dense Tag Identification Using RFID Systems in Vehicular Networks

Xu et al. (2023) conducted a study entitled *Capture-Aware Dense Tag Identification Using RFID Systems in Vehicular Networks*, which focused on improving Radio Frequency Identification (RFID) tag identification in vehicular environments where multiple vehicles and tags are located close to one another, such as at intersections. The study addressed the problem of missed tag identification caused by the capture effect,

which occurs when an RFID reader successfully decodes the signal from the strongest tag while failing to identify other tags transmitting weaker signals. This problem becomes more challenging in dense traffic conditions due to the complexity of wireless communication channels. To address this issue, the researchers established a Nakagami-m distributed channel capture model to analyze capture probability and proposed a capture-aware tag-estimation algorithm to improve the identification of vehicle and cargo tags under different channel conditions.

The researchers evaluated the proposed algorithm through extensive Monte Carlo simulations to determine its performance under fading-channel conditions. The simulation results showed that the proposed approach demonstrated adaptability to different channel conditions and outperformed existing algorithms in terms of computational complexity and tag-identification reliability. The study highlighted the importance of developing efficient identification algorithms to reduce missed tags and improve the performance of RFID systems in environments where multiple tags compete for successful identification.

In contrast, CHARRMPASS focuses on developing a web-based campus entry and exit monitoring system using Radio Frequency Identification (RFID) technology for Aklan State University–Ibajay Campus. While Xu et al. (2023) concentrated on improving tag-identification algorithms and addressing signal interference in dense vehicular networks, CHARRMPASS focuses on identifying individuals and registered vehicles, verifying authorized access, recording entry and exit transactions, and providing centralized monitoring through a web-based dashboard. The system uses low-frequency

RFID cards for individuals and Ultra-High Frequency (UHF) RFID tags for registered vehicles to support campus access monitoring. Its main distinction is the integration of RFID-based identification with web-based record management, visitor management, and report generation in a single system designed for a campus environment. Therefore, CHARRMPASS emphasizes the practical application of RFID technology for organized campus entry and exit monitoring rather than the development and evaluation of tag-identification algorithms for dense vehicular networks.

Radio Frequency Identification(RFID) Based Automated Parking Barrier System

The study focused on improving vehicle access control at the Hasanuddin University Rector's parking area, where security personnel were responsible for manually guarding the entrance and exit gates. This manual process could lead to errors in verifying user authorization and determining which vehicles were permitted to use the parking area. To address these issues, the researchers designed and implemented an automated parking barrier system using Radio Frequency Identification (RFID) technology. The system was intended to automate vehicle authorization and control access to the parking area through RFID card identification.

The system was evaluated based on RFID reading distance, card-reading response time, and the time required to complete the barrier operation. The test results showed that the maximum RFID card reading distance was 9 centimeters when the card was positioned parallel to the reader. The average response time across the four nodes, measured from card presentation until relay activation, was 0.373 seconds. Meanwhile, the total operating times for Nodes 1, 2, 3, and 4 were 6.81 seconds, 18.56 seconds, 11.04

seconds, and 14.13 seconds, respectively. These results demonstrated the system's ability to automate vehicle access control, although the time required to complete the barrier operation varied across the four nodes.

In contrast, CHARRMPASS focuses on developing a web-based campus entry and exit monitoring system for Aklan State University–Ibajay Campus. While the Hasanuddin University study concentrated on automating vehicle access through RFID card verification and parking barriers, CHARRMPASS aims to monitor both individuals and registered vehicles entering and leaving the campus. It integrates RFID-based identification with a centralized web dashboard, digital entry and exit records, visitor management, and report generation. Its main distinction is its broader campus monitoring scope, which supports the management of both individual and vehicle access records rather than focusing primarily on automated parking barriers. Thus, CHARRMPASS applies RFID technology to centralized campus entry and exit monitoring, with an emphasis on access verification and organized digital record management.

RF-Access: Barrier-Free Access Control Systems with UHF Radio Frequency Identification (RFID)

Wang et al. (2022) proposed RF-Access, a barrier-free access control system using Ultra-High Frequency (UHF) Radio Frequency Identification (RFID) technology to address delays and congestion caused by conventional Radio Frequency Identification (RFID) access systems that require users to individually stop and swipe their cards. The study aimed to provide a non-intrusive and contactless authentication process in which

users could pass through an entrance without stopping while the system automatically verifies their Radio Frequency Identification (RFID) tags. The researchers also addressed two major challenges of barrier-free access control: detecting the direction of movement and identifying unauthorized intrusion.

The system integrates a UHF Radio Frequency Identification (RFID) reader, one or more RFID antennas, Radio Frequency Identification (RFID) tags, and a backend server for authentication. As a user passes through the access point, the reader queries the Radio Frequency Identification (RFID) tag and compares its Electronic Product Code (EPC) with authorized identification records stored in the backend server. The system uses a dual-antenna setup and a time-slot-based model to determine the user's direction of movement, while a tag array is used to detect possible illegal intrusion by individuals without Radio Frequency Identification (RFID) tags. Experimental results showed that the prototype achieved 99.83% accuracy in movement direction detection and 96.67% accuracy in illegal intrusion detection, demonstrating the effectiveness of UHF Radio Frequency Identification (RFID) for efficient, contactless, and secure access control.

However, the study primarily focuses on pedestrian access control rather than vehicle identification and automated vehicle entry and exit management. Its barrier-free approach is also different from systems that require a physical gate or barrier to control vehicle access based on Radio Frequency Identification (RFID) authorization. Although the study effectively demonstrates the advantages of Ultra High Frequency (UHF) Radio Frequency Identification (RFID)'s long communication range, high reading rate, backend

authentication, movement-direction detection, and security monitoring, it does not specifically implement these technologies for vehicles or integrate them with an automated vehicle barrier gate.

These limitations show that while the study provides valuable support for the use of UHF Radio Frequency Identification (RFID) in efficient contactless identification and secure access control, further development is needed to apply these capabilities to a web-based vehicle entry and exit monitoring system with automated gate control and real-time tracking, as proposed in CHARRMPASS.

Radio Frequency Identification (RFID): Applications, Security, and Privacy

The study examined the development and research trends of Radio Frequency Identification (RFID) technology, focusing on its applications, security, and privacy. The researchers collected 62,685 records from the Web of Science (WoS) and Scopus databases and processed them to remove duplicates, resulting in 40,677 unique records for analysis. Using the ScientoPy tool, the researchers examined publication and citation indicators, including citation counts, average growth rates, and the number of documents published per year. The review identified major RFID research topics, including Internet of Things (IoT), supply chain management, localization, traceability, logistics, healthcare, access control, authentication, privacy, and ownership transfer. The study aimed to provide an overview of the current research landscape and identify important trends in RFID applications and security.

The study is related to CHARRMPASS because it identifies access control as one of the important application areas of RFID technology. CHARRMPASS also applies RFID for access verification by identifying individuals and registered vehicles entering and leaving Aklan State University–Ibajay Campus. However, while the reviewed study provides a broad analysis of RFID research trends across different applications, security, and privacy topics, CHARRMPASS focuses on developing and implementing a specific web-based campus entry and exit monitoring system. It integrates RFID-based identification with digital access records, centralized web monitoring, visitor management, and report generation. Therefore, the review provides a broader research foundation for understanding RFID applications, particularly in access control, while CHARRMPASS demonstrates the practical application of this technology to campus entry and exit monitoring.

Topic B

Smart Vehicle Access Control System

Smart Vehicle Access Control System refers to an intelligent system that regulates and manages the entry and exit of vehicles in specific areas, properties, or facilities through automated identification, electronic security devices, and centralized management software. It monitors vehicle movements and verifies authorization to ensure that only registered or authorized vehicles are permitted to access designated areas while preventing unauthorized vehicles from entering.

Smart Gateless System Using Radio Frequency Identification (RFID) Technology

Mohd Fazilah et al. (2021) developed a Smart Gateless System using Radio-Frequency Identification (RFID) technology to improve the monitoring and management of vehicles entering and exiting Universiti Malaysia Perlis (UniMAP). The study aimed to address the limitations of conventional campus gate systems by using Radio Frequency Identification (RFID) technology to automatically identify vehicles and record their movements without requiring manual checking by security personnel. The system was designed to provide a more efficient approach to vehicle monitoring while improving campus security and supporting automated access management.

The system integrates Radio Frequency Identification (RFID) technology for vehicle identification and a monitoring system for recording vehicle entry and exit activities. Radio Frequency Identification (RFID) tags are used to identify registered vehicles as they pass through designated monitoring points, allowing the system to capture and process vehicle information automatically.

The collected data is used to monitor vehicle movements and maintain records of campus access, reducing the need for manual inspection. Through this automated identification process, the system supports faster vehicle monitoring and improves the efficiency of managing campus traffic and access.

However, the study primarily focuses on implementing a gateless Radio Frequency Identification (RFID)-based approach for monitoring vehicles within the university environment. Based on the available study details, it does not clearly emphasize an automated physical barrier gate mechanism that can control vehicle access based on Radio Frequency Identification (RFID) authorization. The study also does not fully

describe broader web-based functionalities for centralized administration, such as comprehensive permit management, remote monitoring, and integrated reporting for vehicle entry and exit records. These limitations show that while the study demonstrates the usefulness of Radio Frequency Identification (RFID) technology for automated campus vehicle identification and monitoring, further development can address the integration of Radio Frequency Identification (RFID)-based identification with automated gate control and a centralized web-based platform for real-time vehicle entry and exit monitoring, which are relevant to the proposed CHARRMPASS system.

Enhanced Role-Based Access Control Delegation System

Esomonu and Ikenna (2021) developed an Enhanced Role-Based Access Control Delegation (RBACD) System for tertiary institutions to address difficulties associated with manually controlling access and assigning responsibilities within educational organizations. The study focused on improving access management by assigning roles and permissions to authorized users according to their responsibilities. The researchers proposed an RBAC delegation approach that allows authorized users to delegate tasks while considering the trustworthiness of potential delegates. This approach supports more organized access management and helps ensure that users perform only activities appropriate to their assigned roles. The proposed system applies Role-Based Access Control (RBAC) by associating users with specific roles and permissions. Instead of giving all users the same level of access, the system determines the resources and functions that users can access according to their assigned responsibilities. The delegation mechanism further allows tasks to be transferred to other authorized users when

necessary while considering whether the selected delegate is trustworthy for the assigned task. This approach can reduce the problems associated with manual access control and improve the management of permissions within tertiary institutions.

However, the study primarily focuses on role delegation and access management within tertiary institutions rather than physical campus entry and vehicle access. It does not specifically address RFID-based vehicle identification, automated barrier gates, visitor management, or real-time monitoring of vehicles entering and exiting a university. These limitations show that while the study provides a relevant foundation for assigning permissions according to user roles, further development can apply RBAC to a campus vehicle access environment.

The proposed CHARRMPASS system can use RBAC to assign different permissions to administrators, security personnel, and other authorized users, while combining these roles with Radio Frequency Identification (RFID)-based vehicle authentication and rule-based access decisions. This can help ensure that only authorized personnel can register vehicles, manage access records, monitor vehicle movements, and generate reports, while unauthorized vehicles or visitors can be prevented from entering the campus.

RFID-Integrated Monitoring System

Wahi (2024) developed an Radio Frequency Identification (RFID)-integrated monitoring system at Politeknik Mukah Sarawak to improve the monitoring of individuals entering

and exiting the campus. The study specifically addressed the limitations of manual entry and exit monitoring, which can be time-consuming, prone to errors, and result in inaccurate records.

The proposed system was designed to improve the efficiency, accuracy, and security of campus movement monitoring. The system uses Radio Frequency Identification (RFID) technology and an information-management system to automatically record student movements. Radio Frequency Identification (RFID) credentials are scanned when students pass through the campus access point, allowing the system to record their entry and exit information. The collected information provides the institution with a more systematic record of campus movements and reduces dependence on manually recording individuals entering and leaving the premises.

However, the study focuses on student entry and exit monitoring rather than vehicle access control. It does not specifically implement automated vehicle gates, ANPR, or visitor vehicle permits. Nevertheless, it supports the use of Radio Frequency Identification (RFID) for automated campus entry/exit recording and monitoring, which can be extended in CHARRMPASS to include vehicle identification, visitor authorization, and rule-based access decisions.

Campus Security Access Control System Relying on Association Rules

Qingwei Wang (2021) developed a Campus Security Access Control System Relying on Association Rules at Harbin Huade University, China, to improve campus security and

access management. The study addressed the security concerns of open university campuses, where members of the public may enter and leave campus premises.

The research aimed to use access-control data and association-rule analysis to identify meaningful patterns that could support campus security and help detect potentially abnormal access activities. The system was designed to improve the monitoring of campus personnel and strengthen security decision-making through automated analysis. The system integrates personnel data management, access-data management, and an association-rule mining model for analyzing campus security information. Access records and alarm information are processed and organized into a transaction database, after which association rules are extracted to identify relationships and patterns in the collected data.

The resulting rules can provide meaningful information for campus security personnel and support decisions regarding unusual or potentially concerning access activities. Through this automated rule-analysis process, the system reduces reliance on manually examining large amounts of access information and provides a more systematic approach to campus security monitoring.

However, the study primarily focuses on association-rule analysis of campus access and alarm data rather than directly controlling physical vehicle access. It does not specifically implement Radio Frequency Identification (RFID)-based vehicle identification, automated barrier gates, visitor registration, or a rule engine that immediately grants or denies vehicle entry based on predefined authorization conditions.

These limitations show that while the study demonstrates how rules can be extracted from campus access data to support automated security decisions, further development can apply rule-based logic directly to real-time access authorization.

The proposed CHARRMPASS system can use predefined rules such as whether a vehicle is registered, whether its Radio Frequency Identification (RFID) credential is valid, whether a visitor has an approved permit, and whether the user's role has permission to authorize access. These rules can automatically determine whether entry should be allowed or denied, helping prevent unauthorized vehicles and visitors from entering the campus.

Review of Related Literature

The study focuses on improving campus safety through a door access control system designed for smart campuses. It proposes a two-factor authentication protocol to verify the identities of users and an improved Attribute-Based Access Control (ABAC) model to manage access permissions. The researchers analyze the proposed protocol and evaluate its security and efficiency, reporting that their scheme offers better performance and security compared with other approaches.

Similarly, CHARRMPASS uses Radio Frequency Identification (RFID) technology to support entry and exit monitoring at Aklan State University–Ibajay Campus. It uses low-frequency RFID cards to identify individuals and UHF RFID tags to identify registered vehicles. The system also provides a centralized web-based dashboard where authorized security personnel can verify access and monitor digital entry and exit records.

While the reviewed study emphasizes two-factor authentication and attribute-based access control to strengthen user identity verification and access security, CHARRMPASS focuses on RFID-based identification, access verification, and digital monitoring of individuals and registered vehicles entering and exiting the campus. Both studies aim to improve campus security through technology, but they differ in their primary approaches to access control and monitoring.

Topic C

Real-time Monitoring and Security Dashboard

Real-time monitoring refers to the continuous observation and analysis of systems and data as they operate. A security dashboard provides a centralized visual interface for monitoring threats, tracking incidents, and assessing system status in real time, supporting quick decision-making and proactive security management.

Standalone Radio Frequency Identification (RFID) Access Control System with Data-Integrity Verification Capabilities

Popa et al. (2026) developed a standalone Radio Frequency Identification (RFID) access control system designed to address the limitations of conventional centralized access control systems, particularly their dependence on continuous network connectivity and centralized databases. The system was implemented in a university campus environment and was designed to allow Radio Frequency Identification (RFID)-based access

verification to be performed locally at each access point. Instead of relying on a central database for every access request, authorization information was stored directly on the Radio Frequency Identification (RFID) card and verified by the Access Control Unit (ACU). The system incorporated data-integrity verification and clone-detection mechanisms, where a checksum was generated using the Radio Frequency Identification (RFID) card data and its unique identifier (UID). Any unauthorized modification of the card data resulted in an invalid checksum and caused the system to reject the card. The system also included a graphical user interface (GUI) for administrators to issue and manage Radio Frequency Identification (RFID) credentials, update access permissions, and record administrative activities. In addition, the system was deployed across several university campus buildings and was tested under actual operating conditions for approximately one year.

The study demonstrated strong reliability and security performance during its evaluation. The system completed more than 10,000 authentication attempts, achieving an average authentication latency of 121 ms and a card read failure rate of less than 1%. Security testing resulted in a 100% rejection rate for unauthorized, tampered, expired, and UID-cloneable Radio Frequency Identification (RFID) cards. The decentralized architecture also allowed the access control units to continue operating during network outages because access decisions were performed locally without requiring continuous communication with a central database or the Internet. During approximately 12 months of real-world campus deployment, no critical system failures were reported,

demonstrating the system's ability to maintain reliable access control under actual operating conditions.

However, the study's decentralized and offline-oriented architecture limits its centralized monitoring capabilities. Although the administrative GUI records activities related to Radio Frequency Identification (RFID) card issuance and modification, the study does not provide a centralized web-based dashboard for continuously monitoring vehicle entry and exit logs across multiple campus gates. Furthermore, the authors identified expanded logging and auditing, telemetry, and integration with larger smart campus ecosystems as future work, indicating that its monitoring and data-reporting capabilities could still be enhanced. The system also focuses primarily on building access rather than vehicle traffic monitoring and does not provide analytical traffic reports based on entry and exit patterns.

These limitations show that while the study effectively demonstrates the security, reliability, data-integrity, and offline capabilities of Radio Frequency Identification (RFID)-based access control, further development can combine these advantages with a centralized web-based monitoring platform, real-time vehicle entry and exit logging, and analytical traffic reporting. These aspects are relevant to the proposed CHARMPASS system, particularly in providing reliable access monitoring at campus gates while supporting centralized management and analysis of vehicle access records.

RFID-Based Vehicle Access and Monitoring System

Masri and Ab. Aziz (2025) developed an RFID-based vehicle access and monitoring system for Universiti Tun Hussein Onn Malaysia (UTHM) to address inefficiencies, human errors, traffic congestion, and security risks associated with the university's manual vehicle access procedures. Their system was designed to automate vehicle identification and access authorization using Radio Frequency Identification (RFID) tags, allowing only authorized vehicles to enter the campus while improving the efficiency of vehicle entry and exit procedures. The system incorporated several functions, including user registration, secure login, vehicle and user information management, vehicle access monitoring, unauthorized access alerts, and report generation.

In addition, the system utilized a web-based platform as the primary interface for staff, students, and security personnel, while vehicle information and access records were stored in a centralized MySQL database. The system also provided a security dashboard capable of displaying real-time notifications for unauthorized access attempts and included a reporting and analytics module that provided information on vehicle access patterns and security events.

However, the study identifies statistical analysis as a future improvement, indicating that its analytical capabilities could still be expanded. Furthermore, although the system provides a centralized database, real-time monitoring dashboard, access reporting, and security alerts, it does not specifically address offline operation or local-area-network reliability during network interruptions.

These limitations show that while the study effectively demonstrates the use of Radio Frequency Identification (RFID) and centralized web-based monitoring for campus

vehicle access and security, further development can focus on more comprehensive analytical traffic reporting and reliable operation under connectivity limitations, which are relevant to the proposed CHARMPASS system.

Automated Vehicle Access Control Using License Plate Recognition

The study presents an Automated Vehicle Access Control (AVAC) system developed to address increasing parking demand, access difficulties, and traffic congestion in universities. It uses deep learning-based License Plate Recognition (LPR) to detect vehicle license plates and recognize their characters for vehicle authentication. The system employs the InceptionResNetV2 architecture for license plate detection and a combination of MobileNetV2 and EfficientNetV2 for character recognition. The reported results demonstrate its ability to recognize license plates accurately and process vehicle identification in real time. The system also supports vehicle access control and parking space assignment to improve access management and reduce operational costs.

Similarly, CHARRMPASS is a web-based campus entry and exit monitoring system developed for Aklan State University–Ibajay Campus. Instead of using deep learning-based license plate recognition, it uses UHF RFID tags to identify registered vehicles and record their entry and exit transactions. The system also uses low-frequency RFID cards to identify individuals and provides a centralized web-based dashboard for authorized security personnel to monitor campus access records.

Although both systems aim to improve vehicle access management, they differ in their identification methods and scope. The reviewed study relies on deep learning-based license plate recognition and includes parking space assignment, whereas

CHARRMPASS uses RFID technology for vehicle identification and focuses on monitoring the entry and exit of both individuals and registered vehicles. CHARRMPASS also integrates these records into a centralized web-based system to support campus security operations.

Topic D

Campus Entry and Exit Monitoring

Campus Entry and Exit Monitoring refers to the process of identifying, verifying, and recording individuals and vehicles as they enter and leave a school or university campus. It helps security personnel monitor campus access, determine whether individuals and vehicles are authorized, and maintain organized records of entry and exit activities.

Vehicle Monitoring System with QR Code for the Don Mariano Marcos Memorial State University–North La Union Campus, Bacnotan, La Union

Navalta (2022) developed a Vehicle Monitoring System Using Quick Response (QR) Code for Don Mariano Marcos Memorial State University–North La Union Campus. The study addressed the need to monitor vehicles entering and leaving the campus to support the health and safety of employees, workers, and students. The system was intended to provide a digital method for recording vehicle movements instead of relying entirely on manual monitoring. The researchers used a descriptive-developmental research design and evaluated the system through survey questionnaires involving 20 employees and 20

security guards. The system was assessed according to functional suitability, performance efficiency, usability, reliability, security, and portability. The study provides a relevant example of using QR-based identification for monitoring vehicle entry and exit within a Philippine university setting.

However, the study focused specifically on vehicle monitoring through QR codes and did not employ Radio Frequency Identification (RFID) technology. Its scope was also centered on vehicle entry and exit rather than a broader system that simultaneously accommodates campus users, vehicles, and visitors through different identification mechanisms. This distinction is relevant to the present study because it demonstrates how digital identification can improve vehicle monitoring while leaving room for a more comprehensive campus entry and exit approach.

Web-Based Vehicle Monitoring System of Mindanao State University–Lanao del Norte Agricultural College

Mimbala et al. (2024) developed a web-based vehicle monitoring system for Mindanao State University–Lanao del Norte Agricultural College. The study addressed the campus's existing use of a logbook for recording employees, students, and visitors entering and leaving the campus. The proposed system assigned each registered vehicle a unique QR code that could be scanned at the campus gate. The recorded information was transmitted to a server database, where administrators could retrieve and manage vehicle entry and exit records. The system also incorporated an application programming interface and Short Message Service (SMS) notification for selected personnel.

The study demonstrates the transition from handwritten vehicle logs to computerized entry and exit monitoring. However, it primarily uses QR codes for vehicle identification and does not focus on RFID-based identification of both vehicles and individual campus users. Its vehicle-centered scope provides a different approach from the present study, which considers both campus-user and vehicle identification within the same campus access environment.

Campus Entry and Exit System for Students Using Near Field Communication (NFC)

Mohamed et al. (2025) developed a Campus Entry and Exit System for Students Using Near Field Communication (NFC). The study focuses specifically on recording student movements when entering and leaving the campus. It was developed in response to difficulties associated with manually recording student movement information. The proposed approach uses NFC technology to provide a contactless method of recording student entry and exit activities. The study demonstrates that contactless identification technologies other than RFID can be applied to campus movement monitoring.

The study is relevant because it addresses the same general operational problem of recording student campus entry and exit. However, it uses NFC rather than RFID and focuses primarily on students. It therefore differs from a system designed to accommodate both individual campus users and vehicles using separate RFID credentials.

Prototype of a Facial Recognition System for the Identification of University Students at University Entrances

Núñez and Lino Matamoros (2023) proposed a facial recognition system for identifying university students at institutional entrances. The study was motivated by security concerns and the need to facilitate the identification of authorized individuals entering university facilities. The researchers conducted quantitative field research and gathered data from 20 teachers and 100 students. The results indicated that the respondents supported the development of facial-recognition-based identification because it could help speed up entry and provide an additional security mechanism.

The study demonstrates an alternative approach to campus access monitoring through biometric identification. Unlike RFID systems, facial recognition does not require students to present a physical RFID card. However, facial-recognition systems can be affected by factors such as lighting, facial orientation, and other image-capture conditions. The study is therefore useful for comparing biometric identification with RFID-based identification for campus access.

Validation Intelligently of Automobile and People for Safe and Smart Society

Rastogi, Mishra, and Mishra (2022) developed an automated entry detection system capable of identifying people and automobiles at organizational gates. The system used live video from cameras installed at entry points to recognize registered individuals and record entry and exit activities. Facial recognition was combined with vehicle number-plate detection, while OpenCV, facial-recognition techniques, and a MySQL

database were used to support identification and record management. The system also maintained separate records for human and vehicle entry and exit activities.

The study is highly relevant to campus entry and exit monitoring because it considers both people and vehicles, which is broader than systems that focus only on students or only on vehicles. However, its identification approach depends on camera-based facial and vehicle recognition rather than RFID. The researchers also identified limitations related to image detection, including false detection and recognition issues caused by camera distance. These limitations illustrate the trade-offs between camera-based identification and RFID-based identification.

Synthesis

The reviewed literature and studies show that Radio Frequency Identification (RFID) technology can improve the identification, verification, and monitoring of individuals and vehicles entering and leaving institutional premises. Several studies demonstrated that RFID can reduce dependence on manual recording and checking by automatically capturing identification information and recording access transactions. Poh and Ngadengon (2021), Galicia et al. (2021), Mohd Fazilah et al. (2021), Wahi (2024), and Masri and Ab. Aziz (2025) demonstrated the use of RFID for automated access identification and monitoring, while Wang et al. (2022) and Popa et al. (2026) further showed its potential for contactless access control, security, reliability, and data integrity. These studies establish RFID as a suitable technology for improving the efficiency and security of entry and exit monitoring.

The reviewed studies also show different approaches to campus access monitoring and digital identification. QR-code-based systems were used by Navalta (2022) and Mimbala et al. (2024) to replace handwritten vehicle logs, while Mohamed et al. (2025) used NFC for student entry and exit monitoring. Núñez and Lino Matamoros (2023) explored facial recognition for identifying university students, while Rastogi et al. (2022) combined facial recognition and vehicle number-plate detection to monitor both people and automobiles. These studies demonstrate that different digital and automated technologies can address the limitations of manual campus monitoring. However, they also show that the choice of identification technology depends on the intended users, monitoring environment, and system requirements.

The literature further emphasizes the importance of centralized data management, access control, and monitoring in campus security systems. Esomonu and Ikenna (2021) demonstrated the use of role-based access control for managing user permissions, while Qingwei Wang (2021) showed how access records and rule-based analysis can support campus security monitoring. Masri and Ab. Aziz (2025) integrated RFID identification with a web-based platform, centralized database, monitoring dashboard, alerts, and reporting functions. These findings indicate that an effective monitoring system should not only identify users or vehicles but should also provide organized data management, controlled system access, and accessible monitoring information.

Despite these developments, gaps remain in existing campus entry and exit monitoring systems. Some studies focus only on vehicles, while others focus only on students or pedestrians. Other systems use QR codes, NFC, facial recognition, or license-plate

detection instead of RFID. Some RFID-based studies also focus on a specific entrance prototype, desktop-based monitoring, physical barrier control, or decentralized access verification rather than a centralized web-based monitoring platform. In addition, some systems do not provide comprehensive monitoring of both people and vehicles within the same campus environment. The reviewed studies therefore provide limited integration of RFID-based identification, centralized web-based monitoring, organized entry and exit records, role-based access, and monitoring functions for both individuals and vehicles.

The present study addresses these gaps through the development of a Web-Based Smart Entry and Exit Monitoring System Using RFID Technology for Aklan State University–Ibajay Campus. The proposed system applies RFID-based identification to registered individuals and vehicles, records their entry and exit transactions, and provides a centralized web-based platform for monitoring and managing records. It also incorporates appropriate access permissions for authorized system users and organized reporting of monitoring information. Thus, the reviewed literature and studies provide the technological and conceptual basis for developing a more centralized and integrated approach to campus entry and exit monitoring while addressing the limitations identified in existing systems.

Prior-Arts Comparison Matrix

The Prior-Arts Comparison Matrix presents a comparison between the proposed study and selected existing systems, studies, or related technologies. The matrix identifies the similarities and differences among the proposed study and prior arts based on their

relevant features and functionalities. This comparison helps establish the distinct features, improvements, and contributions of the proposed study.

Table 1. Prior-Arts Comparison Matrix

Features	Proposed Study **CHARRMPASS: A WEB-BASED SMART ENTRY AND EXIT MONITORING SYSTEM USING RADIO FREQUENCY IDENTIFICATION (RFID) TECHNOLOGY**	Prior Art 1CN213582193 (2021)	Prior Art 2IN202131048820 (2022)	Prior Art 3IN202221008803 (2022)

1. Registration and Permit Management	Provides registration of campus user and vehicle information and manages their authorization/permit records.	0	0	0
2. RFID Identification and Access	Uses RFID cards for campus users and RFID stickers/tags for registered vehicles to identify and verify authorized access.	1	0	0
3. Web Dashboard and Monitoring	Provides a centralized web-based dashboard for authorized personnel to monitor entry and exit activities and records.	0	0	0
4. Visitor Management	Provides visitor registration and monitoring for temporary campus access.	0	0	0
5. Emergency Exemption Management	Provides controlled emergency or authorized manual access with	0	0	0

	corresponding records.			
6. Reporting and Records	Maintains entry and exit records and provides monitoring/reporting functions for authorized personnel.	1	1	0
7.Offline Operation and Data Synchronization	Supports operation during connectivity interruptions and synchronizes stored records when connectivity is restored.	0	0	0

CHAPTER III

METHODOLOGY

This chapter presents the research design, system development process locale of the study, respondents of the study, sampling procedure, data-gathering procedure, research instruments, instrument validation, and the statistical tools that were utilized to analyze the data, and ethical consideration.

Research Design

The main purpose of the study was to develop and evaluate CHARRMPASS: A Web-Based Smart Entry and Exit Monitoring System Using Radio Frequency Identification (RFID) Technology designed to address the identified needs and challenges of students, faculty, staff, visitors, and security personnel of Aklan State University–Ibajay Campus, Aklan.

This study utilized a descriptive-developmental research design. The descriptive approach was used to identify and understand the existing entry and exit processes, requirements, needs, and challenges of the intended users, while the developmental approach focused on the systematic design, development, implementation, testing, and evaluation of the proposed system. This research design was appropriate because the study aimed not only to examine the existing access and monitoring processes but also to develop a technological solution based on the identified requirements.

This study utilized the Software Development Life Cycle (SDLC) as the framework for the systematic development of the proposed system. Specifically, the study adopted the Agile model, which provided an iterative and incremental approach to the different stages of system development. The selected SDLC model guided the researchers in identifying requirements, planning the project, designing the system, developing its components, testing the developed functions, deploying the system, and conducting maintenance and evaluation activities.

The Agile model was chosen because it allows the proposed system to be developed iteratively based on identified user requirements and feedback. Through its defined development phases, the researchers were able to progressively develop and

refine the system components while ensuring that the proposed system addressed the identified requirements. The iterative nature of Agile also supported continuous testing, feedback, improvement, and evaluation throughout the development of CHARRMPASS.

System Development Process

The system development process describes the systematic steps followed by the researchers in developing the proposed system. It was guided by the selected Software Development Life Cycle (SDLC) model to ensure that the system was properly planned, developed, tested, implemented, and evaluated.

Phase 1:Requirements Analysis

The first phase focused on identifying and defining the requirements of CHARRMPASS. The researchers examined the existing entry and exit processes at Aklan State University–Ibajay Campus to determine the needs, problems, and requirements of the intended users. Information gathered from the initial assessment, observation, and consultation with relevant users was used to establish the system requirements.

The researchers identified the user requirements, functional requirements, and non-functional requirements of the proposed system. The requirements included registration and permit management, Radio Frequency Identification (RFID)-based identification and access verification, centralized monitoring, entry and exit recording, report generation, visitor management, and emergency vehicle access. The roles and responsibilities of the intended system users were also considered in defining the system requirements.

Phase 2:Planning

The planning phase focused on establishing the scope, resources, development activities, and responsibilities required for the development of CHARRMPASS. The researchers planned the system development based on the requirements identified during the previous phase.

The researchers defined the project scope, required technologies and resources, development schedule, team responsibilities, and project constraints and assumptions. The planned system components and development activities were organized to ensure that the proposed system could be developed according to the identified requirements and within the academic timeframe of the study.

Phase 3:System Design

The system design phase focused on developing the overall structure and design of CHARRMPASS based on the identified requirements. The researchers designed the system architecture, database structure, user interface, and system components necessary for the operation of the proposed system.

The design included the structure of the Registration and Permit Module, Radio Frequency Identification (RFID) Identification and Access Module, Centralized Web Dashboard and Monitoring Module, Visitor Management Module, Emergency Exemption Module, Reporting Module, and Offline Module. The researchers also considered the flow of individual and vehicle information of entry and exit transactions within the system, as well as appropriate security mechanisms for protecting system data and controlling user access.

Phase 4:Development

The development phase focused on the actual implementation of the designed system components. The researchers developed the functional components of CHARRMPASS based on the approved system design and identified requirements.

The development activities included the implementation of the web-based user interface, back-end functions, centralized database, Radio Frequency Identification (RFID) integration, individuals and vehicle registration, access verification, entry and exit recording, monitoring, reporting, visitor management, and emergency exemption functions. The system modules were progressively developed and integrated to form a functional vehicle entry and exit monitoring system.

Phase 5:Testing

The testing phase focused on verifying whether the developed system functions according to the identified requirements and design specifications. Testing was conducted

on the developed components and their integration to identify errors, inconsistencies, and functional issues that required correction.

The researchers conducted appropriate testing procedures, including unit testing, integration testing, and system testing, to verify the functionality and interaction of the system components. The testing activities covered major functions such as individuals and vehicle registration, Radio Frequency Identification (RFID) identification, access verification, entry and exit recording, monitoring, report generation, visitor management, and emergency exemption. Identified issues were addressed and necessary system improvements were incorporated before deployment.

Phase 6:Deployment

The deployment phase focused on preparing the completed system for intended use within Aklan State University–Ibajay Campus. The researchers prepared the system environment, configured the necessary system components, and established the required user accounts and access permissions.

The deployment activities included the installation and configuration of the web-based system and Radio Frequency Identification (RFID) components, preparation of the required system data, and setup of the user access roles. The researchers also provided appropriate orientation to the intended users regarding the basic operation and use of CHARRMPASS.

Phase 7:Maintenance and Evaluation

The maintenance and evaluation phase focused on maintaining the functionality of the developed system and evaluating its quality based on the established evaluation criteria. The researchers reviewed the system for possible errors and areas requiring improvement and incorporated appropriate refinements based on user feedback and identified system issues.

The developed CHARRMPASS system was evaluated using selected characteristics of the ISO/IEC 25010 Software Product Quality Model, specifically Functional Suitability, Performance Efficiency, Reliability, Security, and Maintainability. The evaluation involved selected students, faculty, staff, and security personnel who met the established inclusion criteria. Feedback obtained during the evaluation was considered in identifying areas for system improvement and refinement.

Workplan

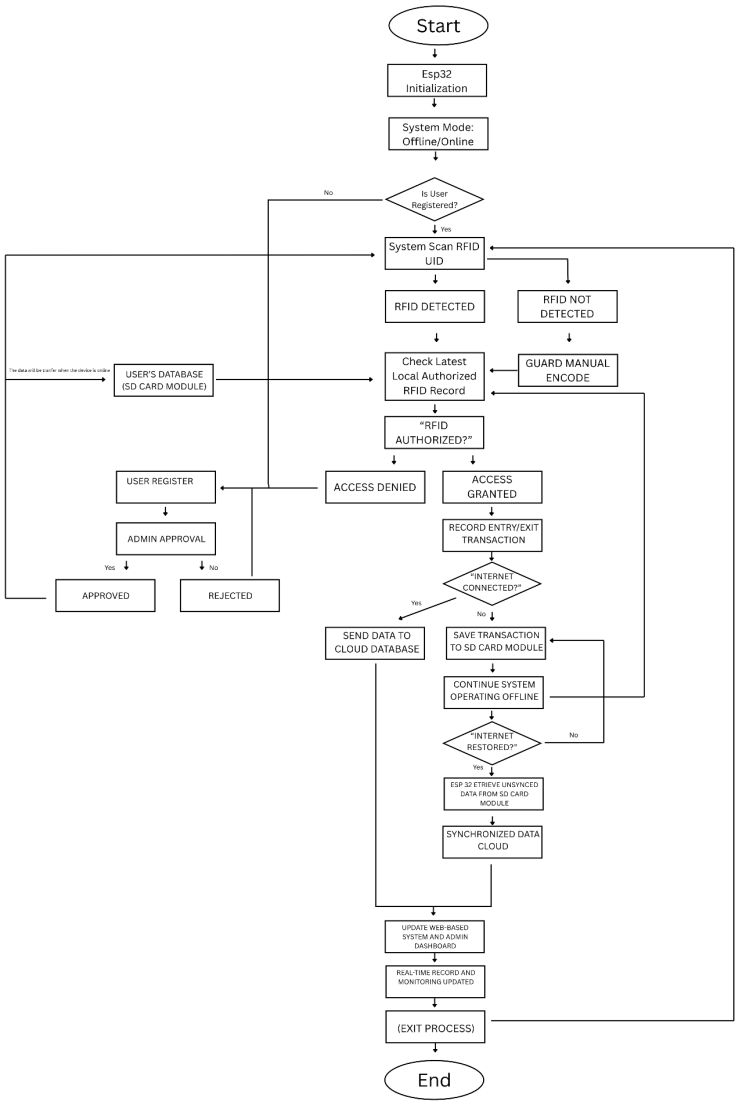
No.	Activities	Weeks Numbers																	
		1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18
	Phase 1: Requirement Gathering Phase																		

	Phase 2: Design the Requirements Phase																	
	Phase 3: Coding (Development) Phase																	
	Phase 4: Testing and Quality Assurance Phase																	
	Phase 5: Deployment Phase																	
	Phase 6: Feedback Phase																	
	Phase 7: Maintenance and Improvement Phase																	

Figure: chcuucuc 1

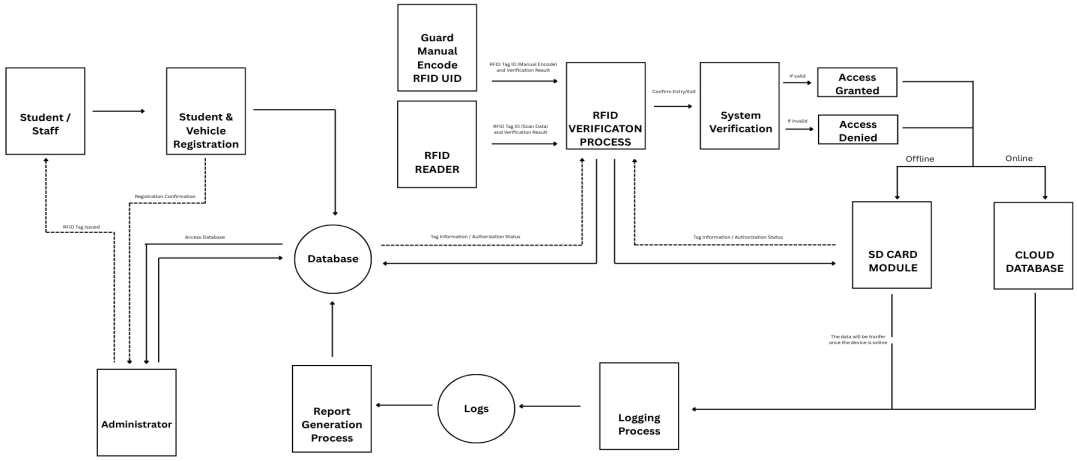
System Flow Diagram

This section presents the system flow diagram of CHARRMPASS: A Web-Based Smart Entry and Exit Monitoring System Using Radio Frequency Identification (RFID) Technology for Aklan State University - Ibajay Campus. The diagram illustrates the step-by-step flow of data and processes within the system, from the entry of individuals and vehicles to their exit, including user interaction, RFID identification, system processing, and output generation.



Data Flow Diagram (DFD)

This section presents the Data Flow Diagram (DFD) of CHARRMPASS: A Web-Based Smart Entry and Exit Monitoring System Using Radio Frequency Identification (RFID) Technology for Aklan State University - Ibajay Campus . The DFD illustrates how data flows between users, system components, and the database, as well as how data is processed, stored, and retrieved within the system.



Prototype (without Code) with link live data

This section presents a prototype of the system without the actual code. The prototype is designed to provide stakeholders with a visual representation of how the system will function. A link to live data is provided to allow stakeholders to view real-time interactions and functionality. Include the QR Code.

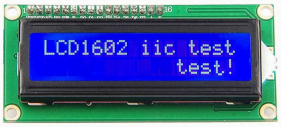


ESP32 - a low-cost, low-power system-on-a-chip microcontroller with built-in Wi-Fi and dual-mode Bluetooth



RFID Card (Radio Frequency Identification Card) - is a contactless plastic card that uses radio waves to transmit

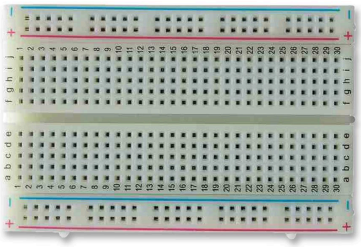
data stored on an internal microchip to a reader without requiring physical contact or a direct line of sight.



LCD Module - is an integrated electronic assembly that combines a liquid crystal display panel with driver circuits, a backlight, and connectors to show images or text.



Jumper wire - is a short electrical wire with pin or socket connectors at each end. It is used to connect two points in a circuit—such as on a breadboard or between a micro-controller like an Arduino and a sensor—without any need for soldering.



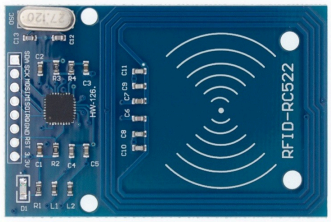
Breadboard - is a plastic tool used to build and test temporary electronic circuits without the need for permanent soldering.



An SD card module is a small electronic adapter board that lets microcontrollers like Arduino read and write data to SD Card

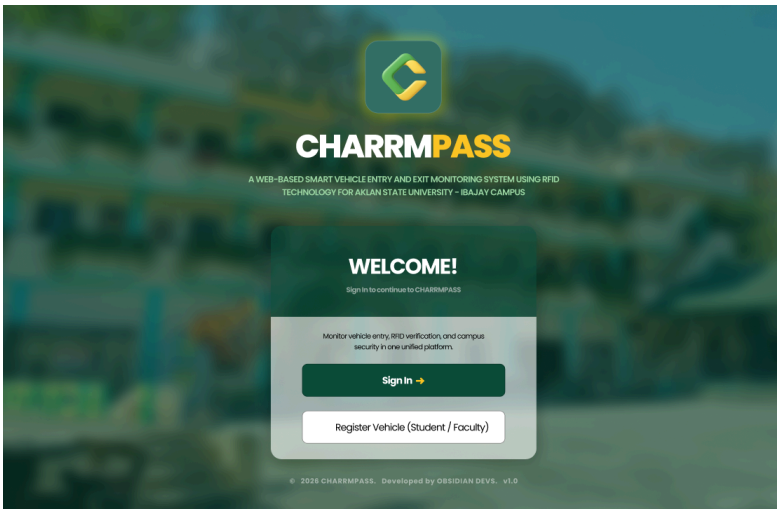


Memory card is a removable Secure Digital High Capacity (SDHC) flash storage device that holds up to 4 gigabytes of digital data.



RFID (Radio Frequency Identification) module scanner is an embedded electronic component that uses radio waves to wirelessly read, and sometimes write, data to tagged items without requiring a direct line of sight

https://www.figma.com/design/OblKCtWP8wmzo6VbqWJj4N/Untitled?node-id=0-1&t=Cl5uJZkNOYYKeQrB-1

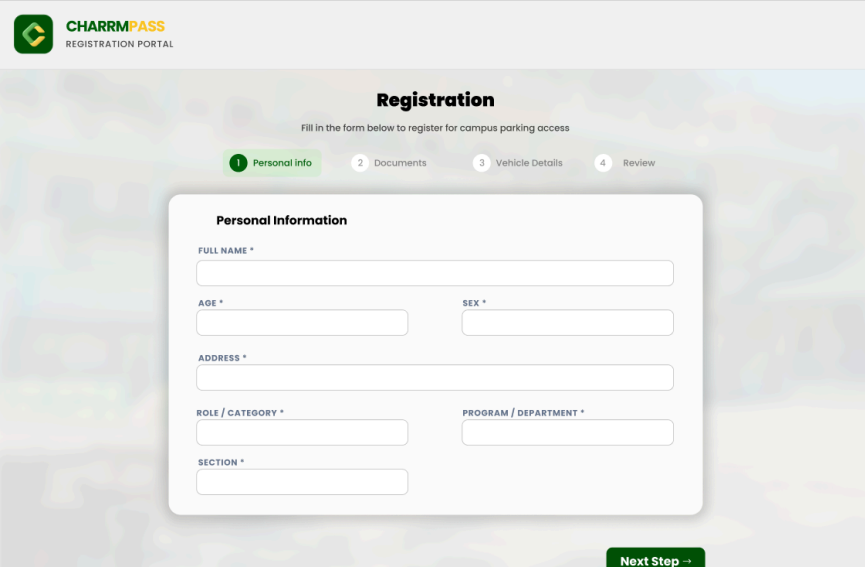


The Welcome/Sign-In Page serves as the initial access point of the CHARRMPASS system. It provides a simple and centralized interface for accessing the system's different user functions. The page is primarily divided into two options: "Sign In" and "Register Vehicle (Student / Faculty)", with each option intended for a specific type of user.

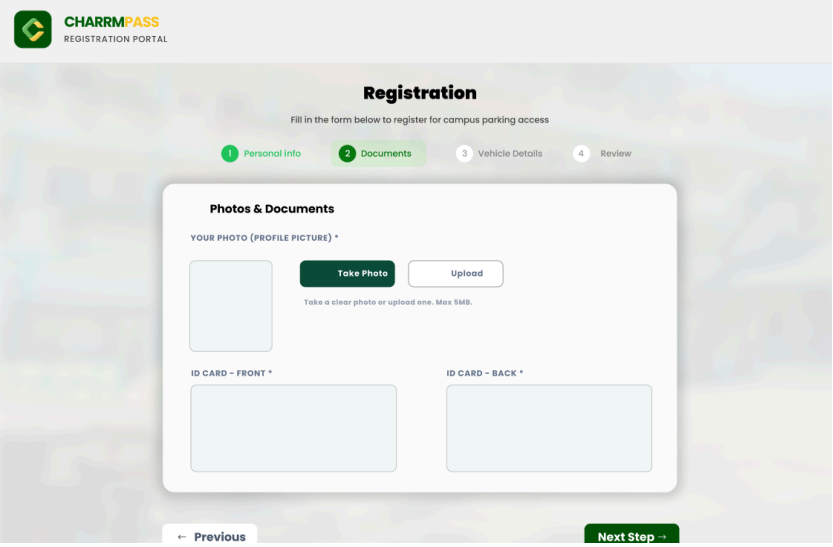
The "Sign In" button is intended exclusively for Guards and Administrators. Each account type has its own username and password and provides access to a separate

dashboard based on the user's assigned role. After entering valid credentials, the system authenticates the account and automatically directs the user to the appropriate dashboard. Guards are directed to the Guard Dashboard for vehicle entry and exit monitoring and Radio Frequency Identification (RFID) verification, while Administrators are directed to the Admin Dashboard for managing system records, registered vehicles, users, monitoring data, and other administrative functions.

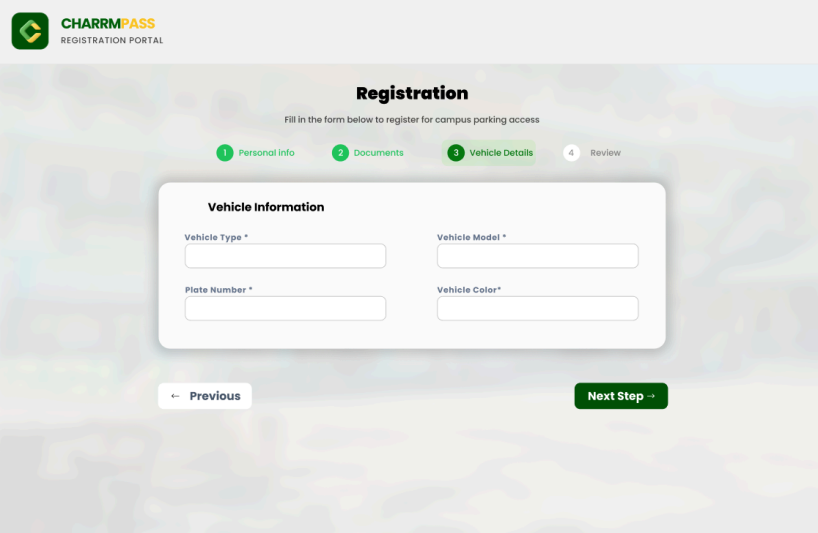
Meanwhile, the "Register Vehicle (Student / Faculty)" option is intended for Students and Staff/Faculty users. These users do not access the Guard or Administrator dashboards. Instead, they can use this option to register their vehicle information in the CHARRMPASS system. This allows their vehicle details to be recorded and associated with their account for proper identification and Radio Frequency Identification (RFID)-based vehicle monitoring.



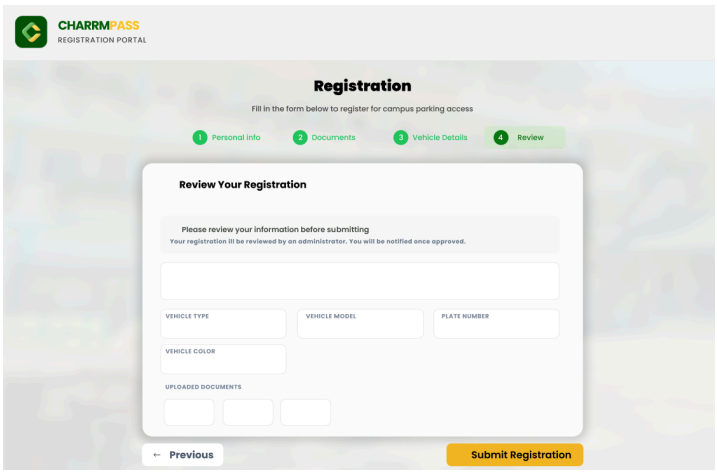
The Registration process begins with the Personal Information interface, where the user enters required personal details such as their name, age, sex, address, course/department, and section. The registration progress indicator shows the current step, and the user can proceed using the “Next Step” button.



After completing the personal information, the user proceeds to the Photos & Documents interface. Here, the user provides the required vehicle photo and ID card images, including the front and back of the ID. The “Previous” and “Next Step” buttons allow the user to navigate between registration steps.

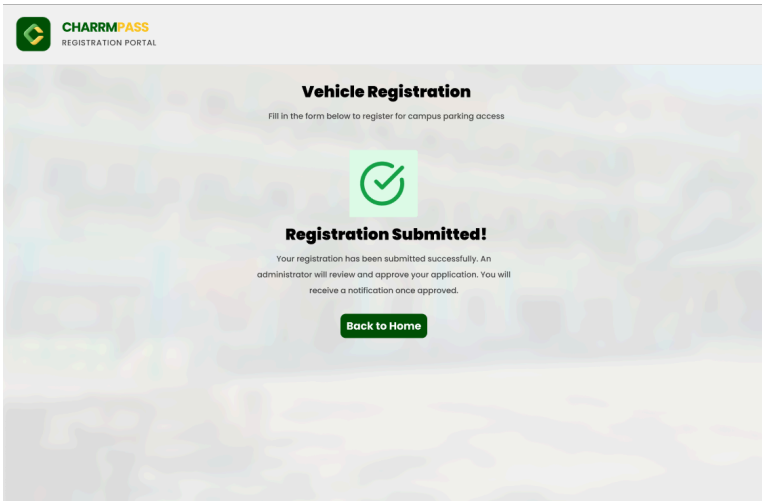


The user then continues to the Vehicle Information interface, where the required vehicle details are entered, including the vehicle type, vehicle model, plate number, and vehicle color. After completing this section, the user can proceed to the next step using the “Next Step” button or return using “Previous.”

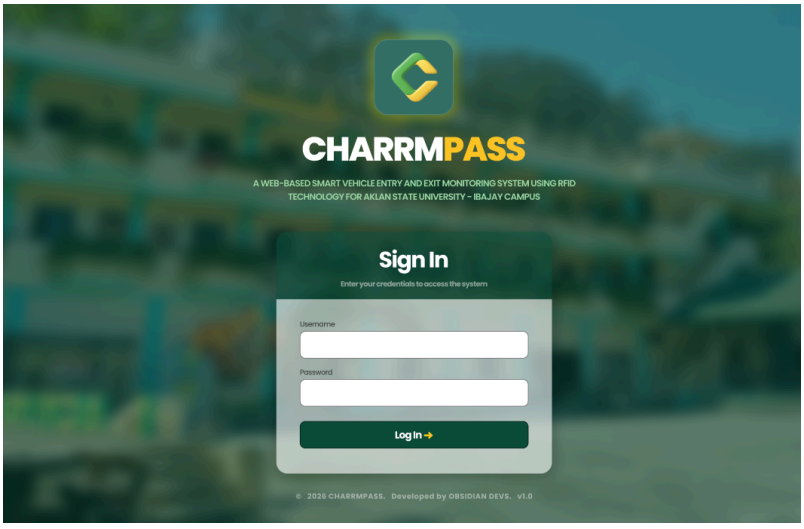


Next, the Review Your Registration interface displays a summary of the information entered in the previous steps. The user can review the registration details

before final submission. The “Previous” button allows corrections, while “Submit Registration” completes the application.



Finally, the Registration Submitted interface confirms that the registration has been successfully submitted. It informs the user that the application has been received for review and provides a “Back to Home” button to return to the main page.

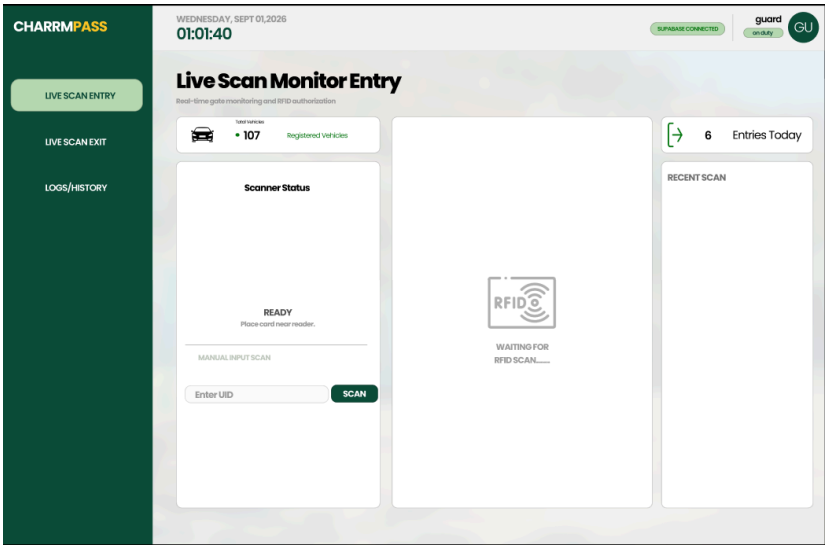


The Sign-In Page serves as the authentication interface for the authorized personnel of the CHARRMPASS system, specifically the Administrator and Guard. It provides a secure login form where users are required to enter their assigned username and password before gaining access to the system.

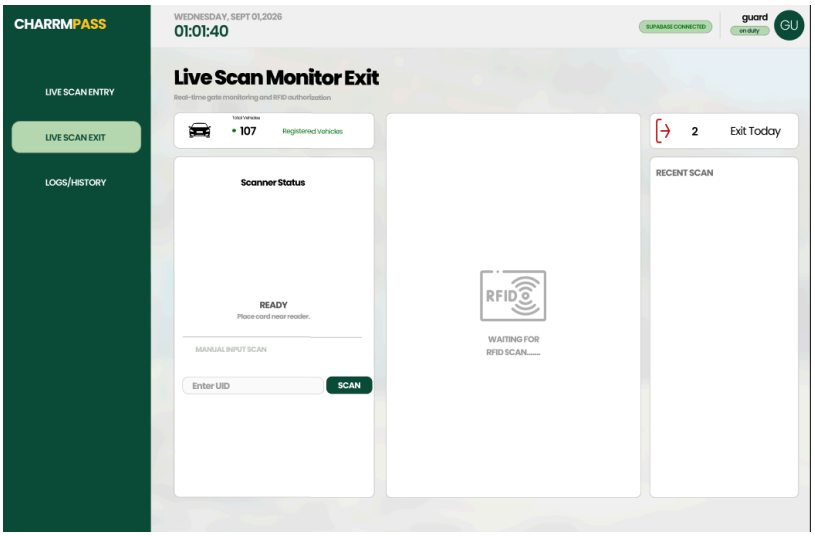
The page contains two input fields for the user's Username and Password, along with a "Log In" button to submit the entered credentials. The system verifies the provided information to ensure that only authorized personnel can access the monitoring and management functions of CHARRMPASS.

After successful authentication, the system identifies whether the account belongs to an Administrator or Guard and automatically redirects the user to their respective dashboard. The Guard is directed to the Guard Dashboard, where they can perform vehicle entry and exit monitoring, Radio Frequency Identification (RFID) verification, and related security functions. Meanwhile, the Administrator is directed to the Admin Dashboard, which provides access to system management, user and vehicle records, monitoring information, logs, and other administrative functions.

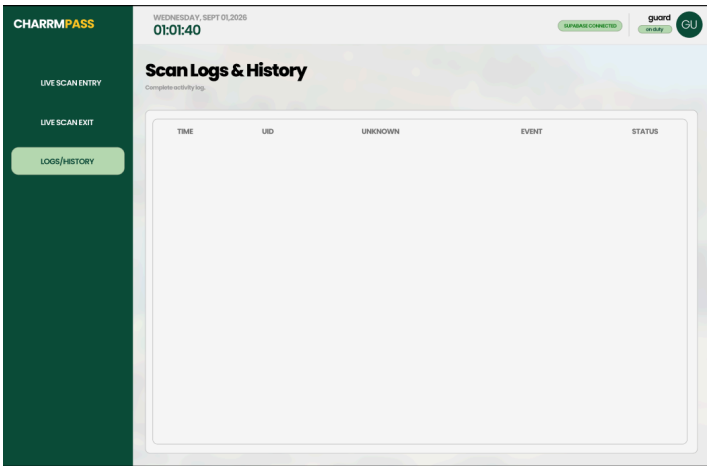
Guard Interface



The **Live Scan Monitor Entry** interface allows Guards to monitor and record **vehicle entries** using Radio Frequency Identification (RFID) technology. It displays the registered vehicle count, daily entry count, scanner status, and recent scans. Guards can scan an Radio Frequency Identification (RFID) tag or manually enter the Radio Frequency Identification (RFID) UID to verify and record an incoming vehicle.

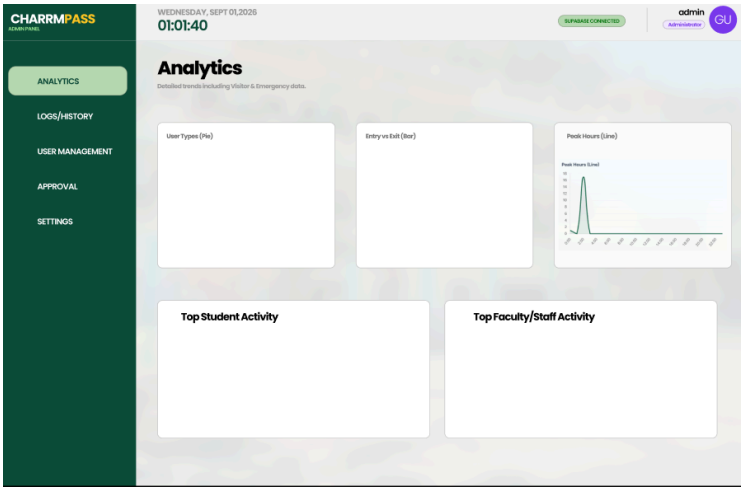


The Scan Logs & History interface allows guards to view recorded Radio Frequency Identification (RFID) transactions. The log displays important information such as the time, Radio Frequency Identification (RFID) UID, vehicle information, event, and status of each scan. This allows security personnel to review and monitor previous vehicle entry and exit records.

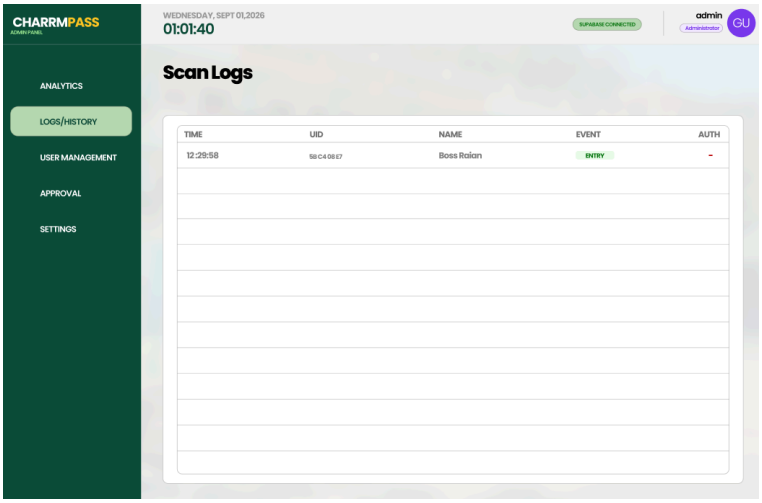


The Scan Logs & History interface allows Guards to view the complete record of vehicle Radio Frequency Identification (RFID) scan activities. It displays important details such as the time, Radio Frequency Identification (RFID) UID, vehicle information, event, and scan status, allowing guards to review and track previous entry and exit transactions.

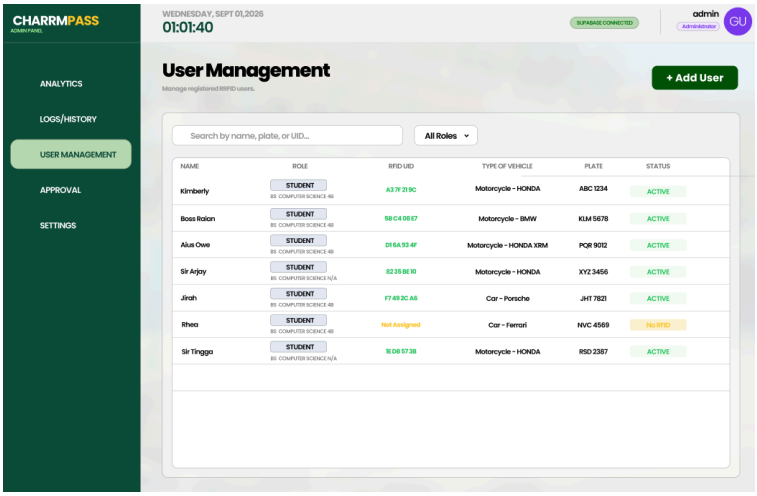
Administrator Interface



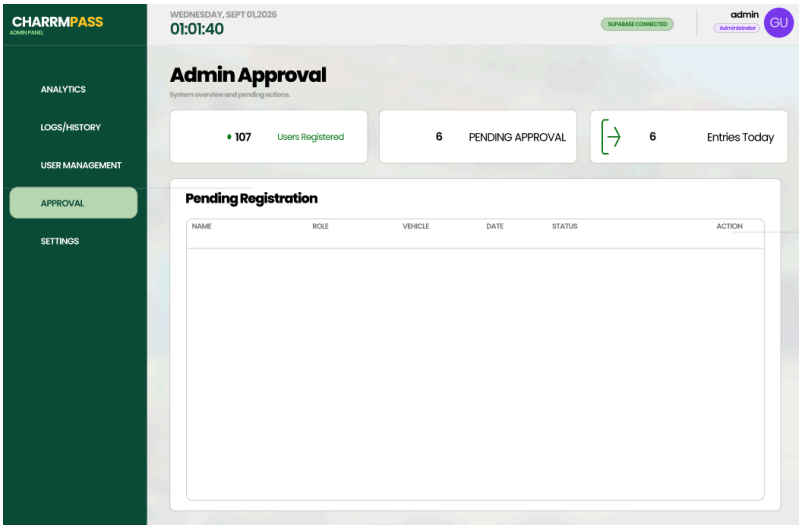
The Analytics dashboard provides an overview of system activity through summary cards and visual data. It displays information such as registered users, vehicle entry and exit activity, peak hours, and the activity of students and faculty/staff. This allows administrators to monitor overall system usage and activity.



The Scan Logs section provides a record of Radio Frequency Identification (RFID) transactions. It displays details such as the time, Radio Frequency Identification (RFID) UID, registered name, event, and authorization status, allowing administrators to review vehicle entry and exit records.

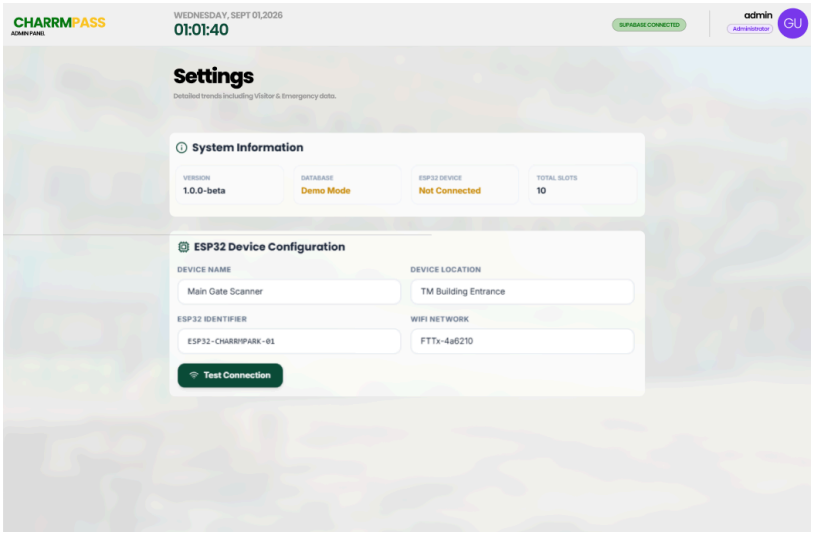


The User Management section allows administrators to manage registered users. It includes a searchable user table containing information such as the user's name, role, Radio Frequency Identification (RFID) UID, type of vehicle, plate number, and account status. An Add User option is also available for adding new users to the system.



The Admin Approval section is used to review pending registrations. It provides summary information such as the number of registered users, pending approvals, and

entries for the day. The Pending Registration table allows administrators to review submitted registrations and take appropriate actions.



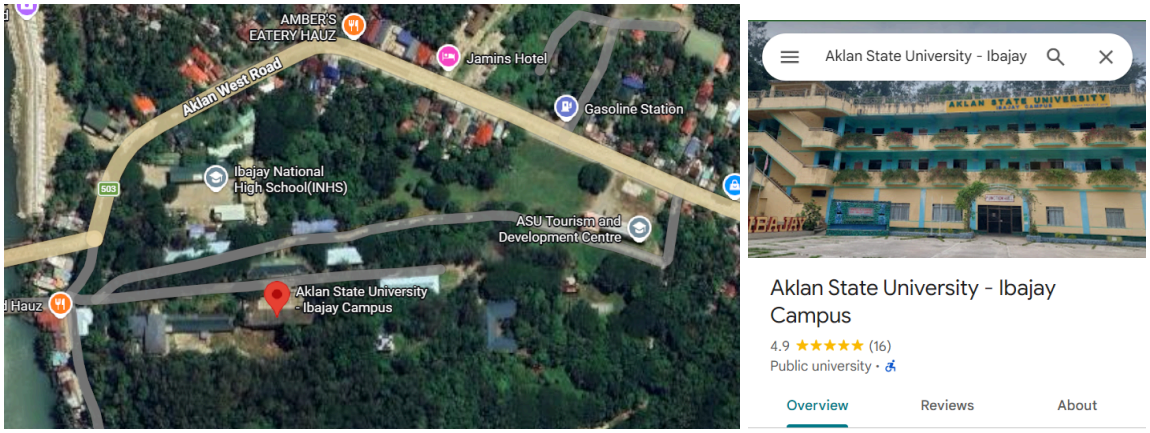
The Settings section provides system and device configuration options. It displays system information and includes an ESP32 device configuration area where the device name, location, ESP32 identifier, and Wi-Fi network can be managed. A Test Connection button is also provided to check the device connection.

Locale of the Study

The study will be conducted at Aklan State University–Ibajay Campus, located in Colong-Colong, Ibajay, Aklan, Western Visayas, Philippines. The locale was selected because it provides the appropriate setting and intended respondents needed for the development and evaluation of CHARRMPASS: A Web-Based Smart Entry and Exit Monitoring System Using Radio Frequency Identification (RFID) Technology for Aklan State University–Ibajay Campus. The campus also provides a suitable environment for

examining entry and exit monitoring processes involving students, faculty, staff, security personnel, and authorized administrators.

Figure 2: *Geographical Location of the Study*



Respondents of the Study

The respondents of the study will be 150 selected students, faculty, staff, and security personnel who will be purposively selected from Aklan State University–Ibajay Campus. They will be selected from a total population of 3,396 eligible students, faculty, staff, and security personnel during A.Y. 2026-2027.

The respondents were selected based on their relevance to the proposed system and their experience or involvement in the processes addressed by the study. Their participation provided the necessary information for identifying system requirements and evaluating the developed system using the selected criteria of the ISO/IEC 25010 Software Product Quality Model.

Table 2. Distribution of Respondents of the Study

Respondent Group	Population	Sample Size	Percentage (%)
Students	**3,211**	**141**	**94.00%**
Faculty	**153**	**7**	**4.67%**
USC Officers	**20**	**1**	**0.67%**
Security Personnel	**12**	**1**	**0.67%**
Total	**3,396**	**150**	**100%**

Sampling Procedure

The proportional allocation sampling technique was used to determine the appropriate number of respondents from each identified group included in the study. This technique ensured that each group was represented in proportion to its population size. The sample size for each group was determined based on its proportion of the total population.

The following formula was used to determine the sample size for each respondent group:

$$n_i = (N_i / N) \times n$$

Where:

- n_i = sample size for each group
- N_i = population of each group

- **N** = total population
- **n** = total sample size

Table 3 presents the distribution of the respondents based on the proportional allocation of the sample. The number of respondents assigned to each group was determined according to the size of its respective population.

Table 3. Distribution of Respondents Based on Proportional Allocation

Respondent Group	**Population**	**Sample Size**
Students	3,211	141
Faculty	153	7
USC Officers	20	1
Security Personnel	12	1
Total	**3,396**	**150**

Data-Gathering Procedure

The data-gathering procedure describes the systematic steps undertaken by the researchers to collect the information needed for the study. First, the researchers will secure the necessary permission from the appropriate authorities of Aklan State University–Ibajay Campus before conducting the study. The researchers will then prepare the data-gathering instruments, including a structured questionnaire and an observation

and assessment guide, based on the objectives, system requirements, and evaluation framework of the study.

After obtaining the necessary approval, the researchers will conduct preliminary data gathering through observation and consultation with selected students, faculty, staff, and security personnel of Aklan State University–Ibajay Campus. The respondents will be selected using purposive sampling based on their relevance to the study and involvement in campus entry and exit activities. The researchers will observe the existing entry and exit procedures and consult the selected respondents to identify the current processes, problems, needs, and requirements related to the identification, verification, recording, and monitoring of individuals and vehicles entering and leaving the campus.

After the preliminary data have been collected, the accomplished instruments and observation records will be reviewed to ensure that the gathered information is complete, relevant, and appropriate for analysis. The information will then be organized, consolidated, and analyzed to identify the requirements of the intended users and the existing entry and exit monitoring process. The identified requirements will serve as the basis for the design and development of the proposed CHARRMPASS: A Web-Based Smart Entry and Exit Monitoring System Using Radio Frequency Identification (RFID) Technology for Aklan State University–Ibajay Campus.

Based on the identified requirements, the researchers will design and develop the system features and modules, including the Registration and Permit Module, RFID Identification and Access Module, Centralized Web Dashboard and Monitoring Module, Visitor Management Module, Reporting Module, Offline Module, and Emergency

Exemption Module. The developed system will undergo testing to verify its functionality and performance before proceeding to the evaluation phase.

After the system has been developed and tested, it will be evaluated by selected students, faculty, staff, and security personnel who meet the established inclusion criteria. The evaluation will use selected characteristics of the ISO/IEC 25010 Software Product Quality Model, specifically Functional Suitability, Performance Efficiency, Reliability, Security, and Maintainability. The evaluation will be conducted using the prepared assessment questionnaire to gather the respondents' feedback regarding the quality and performance of the developed system.

Finally, the collected evaluation data will be tabulated, organized, and analyzed using appropriate descriptive statistical treatments, including frequency, percentage, weighted mean, and standard deviation, as applicable. The results will be interpreted to determine the quality and acceptability of the developed CHARRMPASS system and to identify areas that may require further improvement.

Research Instruments

The researchers will use a researcher-made structured questionnaire as the primary instrument for gathering the necessary data for the study. The instrument will be developed based on the research objectives and the selected criteria of the ISO/IEC 25010 Software Product Quality Model. It will be designed to gather information about the respondents’ assessment of CHARRMPASS and its ability to support RFID-based

identification of individuals and vehicles, access verification, entry and exit recording, centralized monitoring, and visitor management.

The research instrument will consist of three parts. The first part will gather information about the respondents' profiles. The second part will focus on the assessment of the existing entry and exit monitoring processes, including identification, access verification, entry and exit recording, and monitoring of individuals and vehicles. The third part will evaluate the developed system based on the selected criteria of the ISO/IEC 25010 Software Product Quality Model, specifically Functional Suitability, Performance Efficiency, Reliability, Security, and Maintainability.

Before the questionnaire is administered, it will undergo validation and pilot testing to ensure its clarity, relevance, and appropriateness. The pilot testing will involve appropriate participants who will not be included in the final respondents, when feasible. This process will help the researchers identify unclear or ambiguous statements and determine whether the instructions and response options are understandable. After validation and the necessary revisions, the questionnaire will be administered to the selected students, faculty, staff, and security personnel of Aklan State University–Ibajay Campus who meet the established inclusion criteria.

Instrument Validation

The research instrument will be subjected to validation to ensure its validity, clarity, relevance, and appropriateness in gathering the data needed for the study. The instrument

will be reviewed by qualified validators or experts who will be selected based on their expertise in information technology, system development, research, and the subject area relevant to the proposed system.

The validators will examine the instrument in terms of clarity, relevance, organization, consistency, completeness, and appropriateness of the items. Their comments and recommendations will be carefully considered and incorporated by the researchers to improve the instrument before its administration.

After the necessary revisions, the validated instrument will be pilot-tested with appropriate participants who will not be included among the actual respondents, when feasible, to determine whether the questions, instructions, terminology, and response options are clear and understandable. The results and feedback from the pilot test will be considered in finalizing the instrument before it is administered to the actual respondents. This process will help ensure that the instrument is appropriate for collecting reliable and relevant data for the study.

Data Analysis

In this study, the data gathered for the CHARRMPASS: A Web-Based Smart Entry and Exit Monitoring System Using Radio Frequency Identification (RFID) Technology for Aklan State University–Ibajay Campus from the respondents will be organized, tabulated, analyzed, and interpreted using appropriate descriptive statistical methods. The researchers will use frequency, percentage, and weighted mean to analyze

the collected data and determine the results based on the objectives of the study. The responses will be compiled, encoded, and tabulated using Microsoft Excel for data organization, computation, and presentation of the results.

For the respondent profile, frequency and percentage will be used to summarize and describe the distribution of the respondents according to their relevant profile characteristics.

For the evaluation of the developed system, the weighted mean will be used to determine the level of system quality and overall evaluation of CHARRMPASS based on the selected criteria of the ISO/IEC 25010 Software Product Quality Model.

The results were interpreted using the following five-point Likert scale:

Scale	Range	Verbal Interpretation
5	4.21–5.00	Very Satisfied
4	3.41–4.20	Satisfied
3	2.61–3.40	Neutral
2	1.81–2.60	Unsatisfied
1	1.00–1.80	Very Unsatisfied

Table 1. Functional Suitability

Functional Suitability	Score					Weighted Mean	Interpretation
	1	2	3	4	5		
1. **Functional completeness** - Degree to which the set of functions covers all the specified tasks and intended users' objectives.							
2. **Functional correctness** - Degree to which a product or system provides accurate results when used by intended users.							
3. **Functional appropriateness** - Degree to which the functions facilitate the accomplishment of specified tasks and objectives.							
Sub-Mean							**Neutral**

Table 2. Performance Efficiency

Performance Efficiency	Score					Weighted Mean	Interpretation
	1	2	3	4	5		
1. **Time behaviour** - Degree to which the response time and throughput rates of a product or system, when							

performing its functions, meet requirements.							
2. **Resource utilization -** Degree to which the amounts and types of resources used by a product or system, when performing its functions, meet requirements.							
3. **Capacity -** Degree to which the maximum limits of a product or system parameter meet requirements.							
Sub-Mean							**Neutral**

Table 3. Reliability

Reliability	Score						Weighted Mean	Interpretation
	1	2	3	4	5			
1. **Faultlessness -** Degree to which a system, product or component performs specified functions without fault under normal operation.								
2. **Availability -** Degree to which a								

system, product or component is operational and accessible when required for use.								
3. **Fault tolerance -** Degree to which a system, product or component operates as intended despite the presence of hardware or software faults.								
4. **Recoverability -** Degree to which, in the event of an interruption or a failure, a product or system can recover the data directly affected and re-establish the desired state of the system.								
Sub-Mean								**Neutral**

Table 4. Security

Security	Score					Weighted Mean	Interpretation
	1	2	3	4	5		
1. **Confidentiality -** Degree to which a product or system ensures that data are accessible only to those authorized to have access.							
2. **Integrity -** Degree to which a system, product or component ensures that the state of its system and data are protected from unauthorized modification or deletion either by malicious action or computer error.							
3. **Non-repudiation -** Degree to which actions or events can be proven to have taken place so that the events or actions cannot be repudiated later.							
4. Accountability - Degree to which the actions of an entity can be traced uniquely to the entity.							
5. Authenticity - Degree to which the identity of a subject or resource can be proved to be the one claimed.							

6. Resistance - Degree to which the product or system sustains operations while under attack from a malicious actor.							
Sub-Mean							**Neutral**

Table 5. Maintainability

Table 5. Maintainability

Maintainability	Score					Weighted Mean	Interpretation
	1	2	3	4	5		
1. **Modularity -** Degree to which a system or computer program is composed of discrete components such that a change to one component has minimal impact on other components.							
2. **Reusability -** Degree to which a product can be used as an asset in more than one system, or in building other assets.							
3. **Analysability -** Degree of effectiveness and efficiency with which it is possible to assess the impact on a product or system of an intended change to one or more of its parts, to diagnose a product for deficiencies or causes of failures, or to identify parts to be modified.							

4. Modifiability - Degree to which a product or system can be effectively and efficiently modified without introducing defects or degrading existing product quality.							
5. Testability - Degree of effectiveness and efficiency with which test criteria can be established for a system, product or component and tests can be performed to determine whether those criteria have been met.							
Sub-Mean							**Neutral**

The analyzed data will then be interpreted to determine the overall quality and evaluation of the developed CHARRMPASS system and to provide a basis for the conclusions and recommendations of the study.

Ethical Considerations

The researchers observed appropriate ethical principles throughout the conduct of the study to protect the rights, privacy, safety, and welfare of the respondents. The following ethical considerations were observed:

Informed Consent. The respondents were properly informed about the purpose and objectives of the study, the procedures involved, and the nature of their participation.

They were provided with sufficient information about the study and how their responses would be used. Participation was voluntary, and the respondents were asked to provide written or electronic informed consent before participating in the study.

Voluntary Participation. The respondents were informed that their participation was voluntary and without coercion. They were given the right to refuse participation or withdraw from the study at any time without penalty or negative consequences. The researchers respected the decision of each respondent and did not pressure or force anyone to participate.

Confidentiality and Privacy. The researchers ensured that the personal information and responses of the respondents were kept confidential and, when appropriate, anonymized or coded. Information that could identify individual respondents was not disclosed and was used only for academic and research purposes. The researchers also implemented appropriate measures to prevent unauthorized individuals from accessing the collected information.

Data Privacy and Security. The researchers complied with Republic Act No. 10173, or the Data Privacy Act of 2012, by ensuring that personal information collected during the study was collected, processed, stored, and used only for legitimate research purposes. Only information necessary for the study and system evaluation was collected. Personal information related to the evaluation and use of CHARRMPASS, including individual information, vehicle information, and Radio Frequency Identification (RFID)-related identification data, was handled only by authorized individuals and protected against unauthorized access, disclosure, alteration, or misuse.

Beneficence and Non-Maleficence. The researchers ensured that participation in the study did not cause unnecessary physical, psychological, emotional, social, or other risks to the respondents. Data gathering and system evaluation activities were conducted in a manner that promoted the welfare, safety, and well-being of the participants. The researchers also avoided collecting information that was not necessary to achieve the objectives of the study.

Research Integrity. The researchers maintained honesty and integrity throughout the research process. Data were collected, analyzed, interpreted, and presented accurately and objectively. The researchers avoided fabrication, falsification, manipulation, plagiarism, and misrepresentation of data. The findings were reported based on the actual results obtained during the research and system evaluation.

Permission and Authorization. Before conducting the study, the researchers secured the necessary permission and authorization from Aklan State University–Ibajay Campus and the appropriate authorized personnel to conduct the research, gather the necessary information, and involve the identified respondents in the study. The researchers also followed the requirements and procedures established by the institution for conducting research within the campus.

Proper Use of Information. The information gathered during the study was used solely for the purposes of the present research, particularly for the development and evaluation of CHARRMPASS. The collected information was not shared with unauthorized individuals or organizations without appropriate permission or consent. Data were

handled responsibly and retained only for as long as necessary for legitimate research purposes and in accordance with applicable institutional and data privacy requirements.

CHAPTER IV

PRESENTATION, ANALYSIS AND INTERPRETATION OF DATA

This chapter presents the results of the study, the developezd system modules, and the evaluation of the system. The discussion of the results is supported by the findings of related literature and studies reviewed in Chapter 2. The results are analyzed to determine how the developed system addresses the identified problems, needs, and research gaps.

SYSTEM MODULES

This section presents the major modules of the developed _____ *(title/name of the developed system)*. Each module is presented through system evidence, followed by an analysis and discussion relating its features to the findings of previous studies and the research gap identified in Chapter 2.

Module 1: _____*(Title/name of the first system module)*

(Evidence)

(Insert screenshot/screenshot(s) showing the module, interface, features, or actual operation.)

(Discussion)

The _____ *(name of module)* module was developed to _____ *(state the main purpose and function of the module)*. It enables _____ *(identify the intended users)* to _____ *(describe the activities or functions performed by the module)*.

The development of this module is supported by _____ *(author/s and year of related study from the RRL)*, who found that _____ *(briefly state the relevant finding of the study)*. Similarly, _____ *(another author/s and year)* emphasized the importance of _____ *(relevant feature, process, or technology)* in addressing _____ *(related problem)*.

However, the reviewed studies identified a gap in _____ *(specific limitation, missing feature, target user, location, integration, or process not adequately addressed by previous studies)*. The present study addressed this gap by developing _____ *(specific feature/function of the proposed module)*, which provides _____ *(specific improvement, benefit, or contribution)*. Thus, the module demonstrates how the proposed system builds upon previous studies while addressing the identified gap.

Module 2: _____*(Title/name of the second system module)*

(Evidence)

(Insert screenshot/screenshot(s) showing the module, interface, features, or actual operation.)

(Discussion)

The _____ *(name of module)* module was developed to _____ *(state its purpose/function)*. It provides

_____ *(main features)* for _____ *(intended users)*.

This finding is consistent with the study of _____ *(author/s and year)*, which reported that _____ *(relevant finding)*. The study of _____ *(author/s and year)* also showed that _____ *(relevant finding)*.

Despite these developments, previous studies were limited by _____ *(specific research gap)*. The present study addressed this limitation through _____ *(feature/function developed in the module)*. This improvement allows the proposed system to _____ *(specific contribution or benefit)* and responds to the needs identified in the RRL.

Module 3: _____*(Title/name of the third system module)*

(Evidence)

(Insert screenshot/screenshot(s) showing the module, interface, features, or actual operation.)

(Discussion)

The _____ *(name of module)* module was developed to _____ *(purpose/function)*. The module allows _____ *(users)* to _____ *(activities/functions)*.

The findings of _____ *(author/s and year)* support the development of this feature, particularly in terms of _____ *(relevant concept, process, or technology)*. Furthermore, _____ *(author/s and year)* found that _____ *(relevant finding)*.

However, the reviewed literature and studies showed that _____ *(specific gap or limitation)*. To address this gap, the present study incorporated _____ *(specific feature or improvement)* into the developed system. This demonstrates the contribution of the present study in improving or extending the approaches used in previous studies.

Module 4: _____*(Title/name of the fourth system module)*

(Evidence)

(Insert screenshot/screenshot(s) showing the module, interface, features, or actual operation.)

(Discussion)

The _____ *(name of module)* module was developed to _____ *(purpose/function)*. It provides _____ *(main features)* and supports _____ *(specific process or users)*.

The development of this module is related to the findings of _____ *(author/s and year)*, which indicated that

_____ *(relevant finding)*. However, previous studies were found to have limitations in _____ *(specific gap)*.

The present study addressed this gap by incorporating _____ *(specific feature/function)*. This feature provides _____ *(specific improvement or benefit)* and contributes to addressing the identified needs of _____ *(target users/organization/community)*.

SYSTEM EVALUATION

The developed system was evaluated using selected criteria from the ISO/IEC 25010 Software Product Quality Model. The evaluation results were analyzed using _____ *(statistical tool, such as weighted mean)*. The findings were also related to the relevant literature and studies reviewed in Chapter 2 to determine whether the developed system addresses the identified research gap.

Functional Suitability

Table __. Evaluation of the System in Terms of Functional Suitability

(Insert the evaluation table containing the indicators, weighted mean, and verbal interpretation.)

Functional Suitability	Score					Weighted Mean	Interpretation
	1	2	3	4	5		
1. **Functional completeness** - Degree to which the set of functions covers all the specified							

tasks and intended users' objectives.							
2. Functional correctness - Degree to which a product or system provides accurate results when used by intended users.							
3. Functional appropriateness - Degree to which the functions facilitate the accomplishment of specified tasks and objectives.							
Sub-Mean							Neutral

(Interpretation and Discussion)

The table presents the evaluation of the developed system in terms of Functional Suitability. The system obtained an overall weighted mean of _____ *(insert computed weighted mean)*, interpreted as _____ *(insert verbal interpretation)*.

The result indicates that the developed system _____ *(explain what the result means regarding completeness, correctness, and appropriateness of system functions)*. This finding is consistent with _____ *(author/s and year from RRL)*, who reported that _____ *(relevant finding)*.

However, previous studies identified _____ *(specific gap or limitation)*. The present study addressed this gap through _____ *(specific functions/modules/features developed)*.

Therefore, the evaluation result demonstrates that the proposed system provides functional capabilities that respond to the identified needs and extend the functionality identified in previous studies.

Performance Efficiency

Table __. Evaluation of the System in Terms of Performance Efficiency

Performance Efficiency	Score					Weighted Mean	Interpretation
	1	2	3	4	5		
1. **Time behaviour -** Degree to which the response time and throughput rates of a product or system, when performing its functions, meet requirements.							
2. **Resource utilization -** Degree to which the amounts and types of resources used by a product or system, when performing its functions, meet requirements.							
3. **Capacity -** Degree to which the maximum limits of a product or system parameter meet requirements.							
Sub-Mean							Neutral

(Interpretation and Discussion)

The developed system obtained an overall weighted mean of _____ *(insert weighted mean)* for Performance Efficiency, interpreted as _____ *(verbal interpretation)*. This indicates that the system _____ *(explain its response time, resource utilization, or efficiency)*.

The result supports the findings of _____ *(author/s and year)*, which emphasized _____ *(relevant finding from RRL)*. Nevertheless, previous studies were limited by _____ *(specific gap)*. The present study addressed this gap by _____ *(specific technical feature, optimization, or approach)*, thereby improving the efficiency of the proposed system.

Reliability

Table __. Evaluation of the System in Terms of Reliability

Reliability	Score					Weighted Mean	Interpretation
	1	2	3	4	5		
1. **Faultlessness -** Degree to which a system, product or component performs specified functions without fault under normal operation.							
2. **Availability -** Degree to which a system, product or component is operational and accessible when required							

for use.							
3. **Fault tolerance -** Degree to which a system, product or component operates as intended despite the presence of hardware or software faults.							
4. **Recoverability -** Degree to which, in the event of an interruption or a failure, a product or system can recover the data directly affected and re-establish the desired state of the system.							
Sub-Mean							Neutral

(Interpretation and Discussion)

The developed system obtained an overall weighted mean of _____ *(insert weighted mean)* for Reliability, interpreted as _____ *(verbal interpretation)*. This indicates that the system _____ *(explain its consistency, availability, fault tolerance, or recovery capability)*.

This finding is supported by _____ *(author/s and year)*, who stated that _____ *(relevant RRL finding)*. However, the reviewed studies showed a gap in _____ *(specific limitation)*. The present study addressed this gap through _____ *(specific system feature, mechanism, or approach)*,

contributing to a more reliable system for _____ *(target users/organization)*.

Maintainability

Table __. Evaluation of the System in Terms of Maintainability

Maintainability	Score					Weighted Mean	Interpretation
	1	2	3	4	5		
1. <Criterion 1. Description>							
2. <Criterion 2. Description>							
3. <Criterion n. Description>							
Sub-Mean							Neutral

(Interpretation and Discussion)

The developed system obtained an overall weighted mean of _____ *(insert weighted mean)* for Maintainability, interpreted as _____ *(verbal interpretation)*. This indicates that the system _____ *(explain its ease of analysis, modification, testing, or maintenance)*.

The result is related to the findings of _____ *(author/s and year)*, which emphasized _____ *(relevant RRL finding)*. Despite this, previous studies were limited by _____ *(specific gap)*. The present study addressed this limitation by _____ *(specific design, architecture, modular structure,*

documentation, or maintenance feature). Thus, the developed system provides an approach that responds to the identified gap in system maintainability.

CHAPTER V

SUMMARY, CONCLUSIONS AND RECOMMENDATIONS

This chapter presents the summary of the study, conclusions drawn from the findings, and recommendations based on the results of the study.

Summary of Findings

This study focused on the development of _____ *(complete title/name of the proposed system)* for _____ *(target users/organization)* in _____ *(location)*.

The study aimed to _____ *(state the general objective of the study)*. Specifically, it aimed to:

1. Develop the following system modules: _____ *(list all developed modules)*;
2. Evaluate the developed system using the ISO/IEC 25010 Software Product Quality Model, specifically in terms of _____ *(list the selected evaluation criteria)*.

The researchers utilized a _____ *(research design, such as descriptive-developmental research design)* and followed the _____ *(SDLC model, such as Waterfall Model)* in developing the system. The respondents consisted of _____ *(number and type of respondents)* who were selected using _____ *(sampling technique)*. Data were gathered using _____

(research instrument/s used) and analyzed using _____ *(statistical tools used)*.

The developed system consisted of the following major modules: _____ *(Module 1)*, _____ *(Module 2)*, _____ *(Module 3)*, and _____ *(Module 4)*. These modules were designed to address _____ *(specific problems, needs, or gaps identified in the study)*.

Based on the system evaluation, the developed system obtained the following results:

ISO/25010 Criterion	Overall Weighted Mean	Verbal Interpretation
Functional Suitability		
Performance Efficiency		
Reliability		
Maintainability		
Overall Evaluation		

The results showed that the developed system was _____ *(overall result, such as very satisfactory, satisfactory, effective, or acceptable)* based on the selected ISO 25010 criteria.

Conclusions

Based on the findings of the study, the researchers concluded that:

1. The developed _____ *(name/title of system)* successfully addressed _____ *(identified problem, need, or existing process)* through the implementation of the developed system modules. Therefore, the developed system was found to be an appropriate solution for addressing the identified needs and improving the existing process.

2. The _____ *(name of Module 1)*, _____ *(name of Module 2)*, _____ *(name of Module 3)*, and _____ *(name of Module 4)* provided the necessary functions to support _____ *(specific process or activities supported by the system)*. Therefore, the developed modules were considered sufficient in supporting the required functions and activities of the intended users.

3. Based on the evaluation results, the developed system demonstrated _____ *(overall quality, functionality, efficiency, reliability, and/or maintainability)* based on the selected ISO/IEC 25010 criteria. Therefore, the system was found to have an acceptable level of software quality based on the selected evaluation criteria.

4. The overall weighted mean of _____ *(insert overall weighted mean)*, with a verbal interpretation of _____ *(insert verbal interpretation)*, indicates that the developed system _____ *(state what the result demonstrates about the*

system). Therefore, the developed system was considered _____ *(very satisfactory, satisfactory, acceptable, effective, etc.)* based on the evaluation results.

5. The study addressed the identified research gap by _____ *(explain the specific feature, approach, integration, or improvement provided by the proposed system)*. Therefore, the developed system may serve as a _____ *(functional solution, alternative system, management tool, information system, etc.)* for _____ *(target users/organization)* and may contribute to improving _____ *(specific process, service, or activity)*.

Recommendations

Based on the findings and conclusions of the study, the researchers recommend the following:

1. For the Intended Users. The _____ *(target users/organization)* are recommended to _____ *(use, adopt, implement, or integrate the developed system)* to improve _____ *(specific process or activity)*.

2. For the System Administrators. The system administrators are recommended to _____ *(regularly monitor, update, back up, and maintain the system)* to ensure its continuous and reliable operation.

3. For System Enhancement. Future improvements may include _____ *(additional modules, features, integrations, mobile version, notification system, reporting functions, etc.)* to further improve the functionality and usability of the system.

4. For Data Security. The system may be further enhanced by implementing _____ *(additional security measures, access controls, authentication, encryption, backup mechanisms, etc.)* to strengthen the protection of user and system data.

5. For Future Researchers. Future researchers are encouraged to conduct further studies involving _____ *(additional users, larger population, different locations, additional ISO/IEC 25010 criteria, or other related areas)* and to improve the system based on the limitations and findings of the present study.

Areas for Further Research

Considering the limitations of the study, the following are recommended for further verification:

1. Suggestion A
2. Suggestion B
3. …

REFERENCES

Author, A. (Year). Research Title. *Journal Name*, *Volume No.* (Issue No.), pages. DOI.

Please refer to: https://apastyle.apa.org/style-grammar-guidelines

APPENDICES

APPENDICES

This section can include other tables, graphs, pictures, etc. that carry information that is relevant but do not necessarily deserve to be included in the text. Also, include here the curriculum vitae of the student or researcher.

APPENDIX A

Work Breakdown Structure

In this section the researcher will show an illustration on how they formulated and constructed their system successfully.

<Illustration on how your study formulated and constructed successfully>

APPENDIX B

Questionnaire Evaluation Form

A. Profile of Respondents

Name: (Optional) _____ Age: _____ Sex: _____

Designation/Position: _____

Number of years of experience in using computer: _____

B. Evaluation Criteria

Instruction: Please rate the system with the following criteria by checking the appropriate box corresponding your answer based on the rating scale: 1 as the lowest and 5 as the highest.

Interpretation:

1.00 - 1.80 Very Unsatisfied

1.81 - 2.60 Unsatisfied

2.61 - 3.40 Neutral

3.41 - 4.20 Satisfied

4.21 – 5.00 Very Satisfied

Table 1. Discussion

<Criteria 1 Name>	Score					Weighted Mean	Interpretation
1. <Criterion 1. Description>							
2. <Criterion 2. Description>							
3. <Criterion n. Description>							
Sub-Mean							**Neutral**

Table 1. <Name of Table>

Table 2. Discussion

<Criteria 2 Name>	Score					Weighted Mean	Interpretation
1. <Criterion 1. Description>							
2. <Criterion 2. Description>							
3. <Criterion n. Description>							
Sub-Mean							

Table 2. <Name of Table>

Table 3. Discussion

<Criteria 2 Name>	Score					Weighted Mean	Interpretation
1. <Criterion 1. Description>							
2. <Criterion 2. Description>							
3. <Criterion n. Description>							
Sub-Mean							

Table 3. <Name of Table>

Table 3. Discussion

Criteria Name and Sub-Mean	**Sub-Mean**	**Interpretation**
Criteria 1		
Criteria 2		
Criteria n		
Grand-Mean		

Table 4. <Name of Table>

APPENDIX C

Concept Paper Supporting Documents

APPENDIX C.1

Prior Art Search Report

APPENDIX C.2

Technical Problems

APPENDIX C.3

Synthesis

APPENDIX C.4

Proposed Solutions

APPENDIX C.5

Comparison Matrix Between Prior Arts and Proposed Research

APPENDIX C.6

Budgetary Outlay

APPENDIX D

Request Letter to Conduct a Study

In this section the researchers showed the photocopy of the request letter to the end-user to conduct a study approved by the Campus Director.

APPENDIX E

Memorandum of Understanding

In this section, the researchers showed the photocopy of the Memorandum of Understanding between the Researchers, End Users, and School stating their respective duties and responsibilities with proper signatures respectively.

APPENDIX F

Request Letter to Conduct Evaluation of the System(Ongoing/completed Status of the Study)

In this section, the researchers showed the photocopy of the request letter to the end-user to conduct an evaluation of the system such as; functionality, security, and acceptability of the system for the On-going of the study approved by the Campus Director.

<Scanned File>

<Picture 3R Size with Captions)

APPENDIX G

Invitation Letter for the end-user(Ongoing)

In this section, the researcher showed the photocopy of the invitation letter to the end-user to participate in the Pre-Defense as one of the panelists.

<Scanned File>

<Picture 3R Size with Captions)

APPENDIX H

Request Letter to Conduct Evaluation of the System (Completed Status of the Study)

In this section, the researchers showed the photocopy of the request letter to the end-user to conduct the evaluation of the system such as; functionality, security and acceptability of the system for the Completed of the study approved by the Campus Director.

<Scanned File>

<Picture 3R Size with Captions)

APPENDIX I

Invitation Letter for the end-user (Completed)

In this section, the researcher showed the photocopy of the invitation letter to the end-user to participate the Final Defense as one of the panelist.

<Scanned File>

<Picture 3R Size with Captions)

APPENDIX J

Source Code

In this section, the researchers showed the source code of the system.

<Source Code>

APPENDIX K

Grammarian Certification

In this section, the researchers showed the scanned and signed Grammarian certification.

APPENDIX L

Statistician Certification

In this section, the researchers showed the scanned and signed Statistician certification.

APPENDIX K

User's Manual

In this section the researchers will provide a simple and clear step-by-step procedure on how to operate and maintain the system.

APPENDIX L

CURRICULUM VITAE

NAME : <FIRST NAME MIDDLE INITIAL LAST NAME>

ADDRESS : <Complete Address>

MOBILE NO. : <09****>
EMAIL ADDRESS : <*@*.*>

PERSONAL DATA:

COURSE : Bachelor of Science in Computer Science
DATE OF BIRTH : <Month, Day, Year>
PLACE OF BIRTH: <Complete Address>
RELIGION : <Religion>
GENDER : <Male/Female>
CIVIL STATUS : <Single/Married>
HEIGHT : <Feet' Inches">
WEIGHT : <Kilograms>

PARENTS

FATHER : <First Name Middle Initial Last Name>
MOTHER : <First Name Middle Initial Last Name>
HOME ADDRESS : <Complete Address>
DIALECT/LANGUAGES: <Languages>

EDUCATIONAL ATTAINMENT

AKLAN STATE UNIVERSITY <A.Y. Graduated>
College of Hospitality and Rural Resource Management
Ibajay, Aklan
Bachelor of Science in Computer Science

<High School Name> <A.Y. Graduated>
<Address>

<Elementary School> <A.Y. Graduated>
<Address>

SEMINAR AND TRAINING ATTENDED

<Training Title>
<Sponsoring Agency>
<Date>
<Venue>

CURRICULUM VITAE

NAME : <FIRST NAME MIDDLE INITIAL LAST NAME>

ADDRESS : <Complete Address>

MOBILE NO. : <09****>
EMAIL ADDRESS : <*@*.*>

PERSONAL DATA:

COURSE : Bachelor of Science in Computer Science
DATE OF BIRTH : <Month, Day, Year>
PLACE OF BIRTH: <Complete Address>
RELIGION : <Religion>
GENDER : <Male/Female>
CIVIL STATUS : <Single/Married>
HEIGHT : <Feet' Inches">
WEIGHT : <Kilograms>

PARENTS

FATHER : <First Name Middle Initial Last Name>
MOTHER : <First Name Middle Initial Last Name>
HOME ADDRESS : <Complete Address>
DIALECT/LANGUAGES: <Languages>

EDUCATIONAL ATTAINMENT

AKLAN STATE UNIVERSITY <A.Y. Graduated>
College of Hospitality and Rural Resource Management
Ibajay, Aklan
Bachelor of Science in Computer Science

<High School Name> <A.Y. Graduated>
<Address>

<Elementary School> <A.Y. Graduated>
<Address>

SEMINAR AND TRAINING ATTENDED

<Training Title>
<Sponsoring Agency>
<Date>
<Venue>

CURRICULUM VITAE

NAME : <FIRST NAME MIDDLE INITIAL LAST NAME>

ADDRESS : <Complete Address>

MOBILE NO. : <09****>
EMAIL ADDRESS : <*@*.*>

PERSONAL DATA:

COURSE : Bachelor of Science in Computer Science
DATE OF BIRTH : <Month, Day, Year>
PLACE OF BIRTH: <Complete Address>
RELIGION : <Religion>
GENDER : <Male/Female>
CIVIL STATUS : <Single/Married>
HEIGHT : <Feet' Inches">
WEIGHT : <Kilograms>

PARENTS

FATHER : <First Name Middle Initial Last Name>
MOTHER : <First Name Middle Initial Last Name>
HOME ADDRESS : <Complete Address>
DIALECT/LANGUAGES: <Languages>

EDUCATIONAL ATTAINMENT

AKLAN STATE UNIVERSITY <A.Y. Graduated>
College of Hospitality and Rural Resource Management
Ibajay, Aklan
Bachelor of Science in Computer Science

<High School Name> <A.Y. Graduated>
<Address>

<Elementary School> <A.Y. Graduated>
<Address>

SEMINAR AND TRAINING ATTENDED

<Training Title>
<Sponsoring Agency>
<Date>
<Venue>